Mathematics

Year 4

Learner Book

This book was developed with the participation of the Department of Basic Education of South Africa (DBE) using the DBE State-owned (Sasol Inzalo)

FINAL DRAFT COPY

YEAR 4

CONTENTS

TERM 1

UNIT 1: Whole numbers: Counting, ordering, comparing, representing and place value

UNIT 2: Whole numbers: Addition and subtraction

UNIT 3: Whole numbers: Multiplication and Division

UNIT 4: Ratio and rate

UNIT 5: Common fractions

UNIT 6: Decimal fractions

UNIT 7: Whole numbers: Finance

TERM 2

UNIT 1: Whole numbers: Counting, ordering, comparing, representing and place value

UNIT 2: Whole numbers: Finance

UNIT 3: Algebraic Language

UNIT 4: Numeric and Geometric Patterns

UNIT 5: Measurement

UNIT 6: Construction of Geometric Figures

TERM 3

UNIT 1: Geometry of straight lines

UNIT 2: Geometry of 2-D shapes

UNIT 3: Transformations

UNIT 4: Geometry of 3-D Objects

UNIT 5: Area and Perimeter

UNIT 6: Surface area and Volume

UNIT 7: Data Handling

TERM 4

FINAL DRAFT COPY

UNIT 1: Revision of term 1 and 2 work

- Whole numbers
- Fractions
- Algebraic Language
- Numeric and Geometric patterns

UNIT 2: Revision of term 2 and 3 work

- Measurement
- Construction of Geometric Figures
- Geometry of Straight Lines
- Geometry of 2-D Shapes
- Transformations
- Geometry of 3-D Objects

UNIT 3: External Moderation of SBA and Final External Pen on Paper Assessment

FINAL DRAFT COPY

TERM 1

FINAL DRAFT COPY

FINAL DRAFT COPY

FINAL DRAFT COPY

UNIT 1

WHOLE NUMBERS: COUNTING, ORDERING, COMPARING, REPRESENTING AND PLACE VALUE

1.1 Counting, ordering, comparing and representing to at least 8 - digit whole numbers

1. Fill in the next number:
   (a) 12 200 000; 15 200 000; 18 200 000; -------
   (b) 48 500 750; 48 500 500; 48 500 250; -------
   (c) 91 400 000; 88 250 000; 85 100 000; -------
   (d) 22 444 111; 44 444 111; 66 444 111; -------

2. Copy and complete the table. Count in 55 000s

19 000 000	(a)	19 110 000
19 055 000	19 110 000	(b)
(c)	(d)	19 220 000

Order, compare and represent whole numbers

3. Arrange the following number in ascending order.
   (a) 27 153 999; 27 135 999; 27 315 999; 27 513 999
   (b) 50 101 101; 50 010 101; 50 100 110; 50 001 011

4. Arrange the following number in descending order.
   (a) 33 923 673; 33 926 373; 33 923 763; 33 926 337
   (b) 41 561 152; 41 516 512; 41 615 215; 41 651 125

Comparing whole numbers

5. Compare the underlined digits and explain the relationship that you see in your own words.
   (a) 56 10**1** 101 and 65 101 10**1**
   (b) 97 1**7**2 851 and 9**7** 271 158

FINAL DRAFT COPY

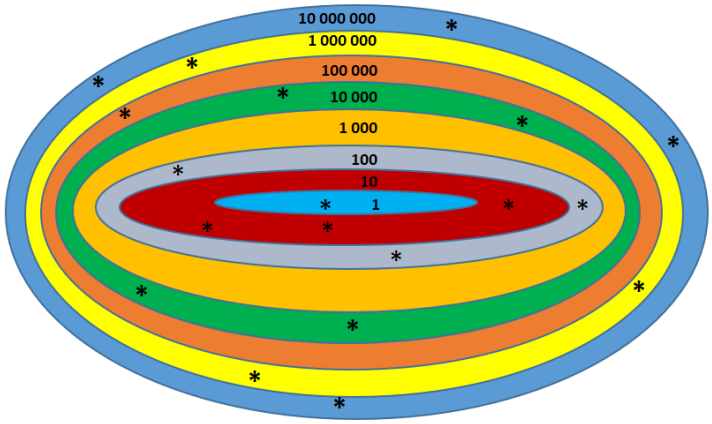
6. Compare these numbers by inserting $<; > or =$

(a) 90 299 992 ---- 90 292 999

(b) 11 723 723 ----11 723 723

Representing whole numbers

7. Study the following diagram and answer the questions:



(a) You have dropped some stars onto a game board. This was the result. If you add the numbers what is the total?

8. You have been given these eight digits:

1	7	9	3	0	2	4	5

(a) Use each digit once. Make the smallest 8 - digit whole number

(b) Use each digit once. Make the largest 8- digit whole number

(c) You can use a digit twice. Make the largest 8 - whole number

(d) You can use a digit twice. Make the smallest 8 - digit whole number

1.2 Recognise the place value of whole numbers to at least 8-digits

1. Write the number(digit) in the correct column

No.	Number	TM	M	Hth	Tth	T	H	T	O
(a)	27 683 005								
(b)	80 765 011								

FINAL DRAFT COPY

2. Write the number symbol for each number.
   (a) Thirty million five thousand
   (b) Thirty million and five
   (c) Thirty million fifty thousand

1.3 Rounding off to the nearest 10, 100, 1 000, 10 000, 100 000 and 1 000 000 to estimate answers.

1. Round off each of the numbers as indicated in the following table.

Number	Nearest 10	Nearest 1 000	Nearest 10 000	Nearest 100 000
23 400 136				
45 555 555				

2. The first number below was rounded off to the second number. Was it rounded off to the nearest 10, 100, 1000, 10 000,100 000 or 1 000 000?
   (a) R1 676 241 was rounded off to R1 676 200
   (b) R2 575 001 was rounded off to R3 000 000

1.4 Solving problems in context involving counting, ordering, comparing, representing, place value and rounding off.

1. My brother remembers seeing an affordable spacious block of flats for sale. He could not remember the exact price, but he knew it was an eight-digit number. He remembered that the last digit was a five and in the hundred thousand place was a three. He remembered that the first digit was one and the ten thousand place was four. In the millions place he remembered the number was two and, in the tens, place the number was three times three. Finally, he said the number nine was in the hundreds place. What is the number?
2. A baker needs to have an estimate of how many biscuits are baked per week in her bakery. She normally rounds off the tallies and then adds them. The frequencies are as follows: 7 653 492; 4 892 222; 6 600 154.
   Round off these tallies to the nearest million and add them.

FINAL DRAFT COPY

3. You live in a street with 6 houses. Each family owns a car. The first family's car costs two hundred thousand rands. The second family's car costs forty-five thousand less than the first family's car. The third family's car costs double the first family's car. The fourth family a half million rands for their car. The fifth family paid fifty thousand rand more than the fourth family, and the last family paid twenty thousand rands less than the third family.
   Calculate the price of each car and arrange the prices in descending order?
4. I am greater than seventy but less than ninety, the sum of my two digits is thirteen, my one's digit is a multiple of two. What number am I?

FINAL DRAFT COPY

UNIT 2

WHOLE NUMBERS: ADDITION AND SUBTRACTION

2.1 Use a range of techniques to perform written calculations with whole number

Addition and subtraction in columns

Addition	Subtraction
21 768 421 + 16 593 094 =	23 473 546 – 22 181 362
21 768 421 + 16 593 094 **38 361 515**	23 473 546 – 22 181 362 **1 292 184**

1. Calculate the following using the column method. Use your calculator to check your answers.
   (a) 763 264 + 39 652
   (b) 19 054 947 + 11 593 204
   (c) 798 456 – 573 245
   (d) 482 563 – 356 278
   (e) 2 839 632 – 1 573 245
   (f) 31 695 481 + 22 397 447 – 7 356 278

Adding and subtracting using a calculator

The basics of using a calculator are the same whether one is using a standard or scientific calculator. Once one learns what the buttons do and how to use them for different calculations, one becomes more comfortable using a calculator whenever necessary. Remember to clear the calculator memory before starting a new problem.

a new problem.

Example 1

Keystroke sequence gives a guide on the exact keys to press

The calculator plan

124 786 + 639 536

Calculator keystroke sequence

1	2	4	7	8	6	+	6	3	9	5	3	6	=

FINAL DRAFT COPY

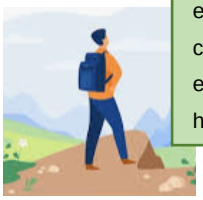
757 869 – 295 425 7 5 7 8 6 9 – 2 9 5 4 2 5 =	

2. Calculate each of the following using the column method and check your answers using a calculator.
   (a) 15 987 364 + 49 790 683
   (b) 76 263 785 + 57 953 064
   (c) 87 623 735 – 37 470 184
   (d) 97 654 231 – 74 933 923
   (e) 31 511 547 + 14 923 789 – 21 573 245

2.2 Problem solving involving addition and subtraction

1. The people in a village get their water from a nearby dam. On a certain day the dam contains 688 000 litres of water. The village people use about 595 000 litres of water per week.
   (a) Will the community have enough water for the following week if no rains fall?
   (b) Is it possible to calculate the exact amount of water that will be in the dam after a week?

2. Moses went hiking in the mountains for 6 days. He started on Monday and walked 6 kilometers and his last trip was on Saturday. Each day he walked one kilometer more than the day before. What will be the total distance walked after 6 days?


https://www.freepik.com/vectors/hiking

3. A bank manager is working on the budget for a year. He has to try to keep the total expenditure on new office equipment below R800 000. He still has to budget for new computers that are badly needed, but this is what he has written so far:

ITEMS	COST
62 × new office chairs	R 54 020,00
38 × new computer screens	R 100 800,00
11 × new printers	R 141 600,00
15 × new tea trolleys	R 25 740,00

FINAL DRAFT COPY

10 × new carpets for senior staff offices	R 144 000,00
120 × small plastic filing cabinets	R 52 380,00
1 × new table for the boardroom	R 48 000,00
18 × new chairs for the boardroom	R 41 400,00

(a) How much has the bank manager budgeted for printers and computer screens together?

(b) How much, in total, has the bank manager budgeted for chairs and tables?

(c) Work out the total cost of all the items the municipal manager has budgeted for.?

(d) Can the bank manager add new computers for 3 senior staff members if each computer costs R30 000,00?

4. Benjamin had the following expenses last month:

Movie R30; Bus R280; Donation to welfare R50; Stationery R65; Repairs to bicycle R225; New shirt R135; Ice cream R23. He receives R800 pocket money per month. He has R325 saved thus far.

Complete the budget to find out if he will have enough money or he will need to tap on his savings.

Item	Estimated amount	Actual amount	Difference
Income	R800		
Expenses			
Transport	R200		
Stationery	R100		
Clothes	R100		
Food	R100		
Entertainment	R50		
Donations	R50		
Savings	R100		
Emergencies	R100		
Estimated total Expenses			
Net income			

FINAL DRAFT COPY

UNIT 3

WHOLE NUMBERS: MULTIPLICATION AND DIVISION

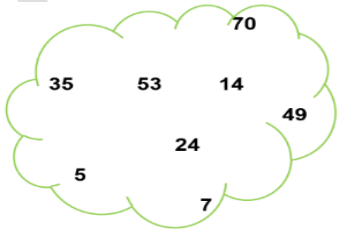
3.1 Use a range of techniques to perform written calculations with whole numbers

Multiples and Factors

1. Complete the following table

Number	Factors	First three multiples
101		
285		
136		

2. Answer the questions based on the group of numbers given in the diagram.



(a) List the factors of 35

(b) List the multiples of 7

(c) Which numbers are multiples of 14?

(d) Which numbers have 5 as a factor?

3. Write down the answers as quickly as you can.

(a) 4 355 × 1

(b) 7 227 × 0

(c) 0 + 3 000

(d) 5 653 + 0

4. Multiply and check the answers using a calculator.

(a) 1 117 × 522

(b) 3 641 × 183

(c) 2 954 × 300

(d) 5 324 × 219

5. Divide and check the answers using a calculator.

(a) 6 478 ÷ 111

(b) 5 079 ÷ 263

FINAL DRAFT COPY

(c) 9 864 ÷ 274

(d) 7 822 ÷ 320

3.2 Problem solving involving division and multiplication

1. Mrs. Nkosi is buying apples from the market and sell them to school children. Last weekend she bought 3 boxes of apples; each box contained 15 packets of 12 apples each. Each box costs her R180,00. She managed to sell 80% of the apples at R1,20 before the rest were eaten by her grandchildren and their friends. Did she make a profit or a loss?
2. Sihle is buying pens for R5,80 each and he wants to make a 50% profit. How much must he sell them for?
3. The Minister of Mineral Resources and Energy announced that the price of petrol is going to decrease by 20%. If your budget was R4 500 a month for petrol, how much will you budget now?
4. My father wanted to purchase a laptop he saw online. When he made the offer to purchase, they told him that the price does not include VAT. The original price was R8 995. How much will the laptop cost including VAT?

FINAL DRAFT COPY

UNIT 4
RATIO AND RATE

Ratio is a way of comparing two or more quantities of the same kind. It is a way of comparing one part of a whole with another part of the same whole.
Rate is a way of comparing two different quantities. It is used to indicate how much of a given quantity is associated with one unit of another quantity.

Revision Activity

1. A school tap is licking. 5 drops of water are lost per minute. How many drops will be lost in 15 minutes?
2. Sihle has R32 and Ntokozo has R4. Write down the ratio of Sihle's money to Ntokozo's money.
3. If 3,5kg of bananas cost R12, what will 1000g of bananas cost?
4. Peter and James share 28 sweets in the ratio of 3:1. How many sweets will each one get?

4.1 Ratio in the context of speed, distance and time

Examples

Mpho collects apples in the orchard. She picks about 5 apples each minute. Approximately how many apples will Mpho pick in each of the following periods of time?

- 8 minutes — 8 min x 5 apples = 40 apples
- 11 minutes — 11 min x 5 apples = 55 apples
- 15 minutes — 15 min x 5 apples = 75 apples
- 20 minutes — 20 min x 5 apples = 100 apples

Musa and Karabo also collect apples in the orchard, but they both work faster than Mpho. Musa collects apples at a rate of about 12 apples per minute and Karabo at a rate of about 15 apples per minute.

FINAL DRAFT COPY

- Complete the following table to show approximately how many apples they will each collect in different periods of time.

Time in minutes	1	2	3	8	10	20
Mpho	5					
Musa	12					
Karabo	15					
The Three together	32					

The number of apples picked is directly proportional to the time taken.
If you filled the table correctly, you will notice that during any period of time, Karabo collected 3 times as many apples as Mpho. We can say that during any time interval, the **ratio** between the numbers of apples collected by Mpho and Karabo is **three is to one**, which can be written as **3:1**. For any period of time, the ratio between the number of apples collected by Musa and Mpho is **12:5.** Ratio should always be given in its simplest form.

1. What is the ratio between the number of apples collected by Karabo and Musa in two minutes?
2. What is the ratio between the number of apples collected by Karabo and Mpho in three minutes? Write the ratio in simplest form
3. How much will each person get, if R14 400 is shared between two people in each of the following ways?
   (a) In the ratio 1: 3
   (b) In the ratio 5: 7
4. R240 will be divided between Amahle and Mandy in the ratio 3: 5. This means Mandy gets R5 for every R3 Amahle gets. How much will Amahle and Mandy each get in total?.
5. Norman, Sello and Thabani worked in Mr Manzi's garden. Norman worked for 5 hours, Sello for 4 hours and Thabani for 3 hours. In total the Mr Manzi paid the boys R600. If the money was divided in the ratio of 5:4:3, how much did each boy get?

FINAL DRAFT

4.2 Rate in the context of speed, distance and time

Example

Michael drives from Mabeskraal in Rustenburg to Mbombela in Nelspruit at an average speed of 88 km/h. The distance between these two towns is 440 km. How long did it take Michael to get to Mbombela?

Solution

Speed = distance ÷ time

∴ Time = distance ÷ speed

= 440 ÷ 88

= 5 hours

It took Michael 5 hours to get to Mbombela.

1. A motorist covers a distance of 425 km in exactly 5 hours
   (a) Approximately how far did the motorist drive in 1 hour?
   (b) Approximately how far will the motorist drive in 9 hours?
   (c) Approximately how long will the motorist need to travel 900km?
2. Mary ran a 20km race. Her average speed was 8km per hour. How long did it take her to complete the race?
3. If a car uses 7,5 litres of petrol for every 100 km that it travels, how many litres would it use for a trip of 620 km?
4. It takes 20 minutes for 10 teachers to mark 20 scripts. How long will it take 5 teachers to mark 10 scripts?
5. The marathon runner runs 16 km in 35 minutes. How long will it take him to walk the same distance if he walks 4 km in 12 minutes?
6. A car travels 46 km in 16 minutes. At the same average speed, how far will it travel in 1hour 12 minutes?
7. If it takes 7 people to make 21 T-shirt per day, how long will it take 14 people?

FINAL DRAFT COPY

UNIT 5

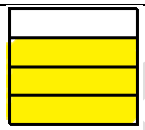
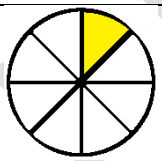
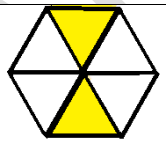
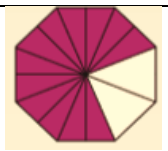
COMMON FRACTIONS

Revision

1. Write the following as common fractions and simplify where possible

1.	1 ÷ 12		2.	2 ÷ 8	
3.	11 ÷ 10		4.	5 ÷ 9	
5.	5: 9		6.	12: 18	
7.	15:12		8.	25%	
9.	0,2		10.	1,5	

2. Copy and complete the table below.

No	Diagrammatic form	Name	Symbol	3 equivalent fractions
(a)		One third or a third	$\frac{1}{3}$	$\frac{3}{9} = \frac{2}{6} = \frac{10}{30}$
(b)				
(c)				
(d)				
(e)				
(f)				

FINAL DRAFT COPY

3. Say whether the following are **true** or **false**. Show how you arrived at an answer.

(a) $1 = \frac{2}{2} = \frac{5}{5}$

(b) $\frac{1}{3} = \frac{1}{6} + \frac{1}{6}$

(c) $5{,}6 = 5\frac{6}{10}$

(d) $20\% = \frac{1}{4}$

(e) $\frac{3}{8} + \frac{1}{8} = \frac{1}{4}$

5.1 Compare and order common fractions, including tenths and hundredths

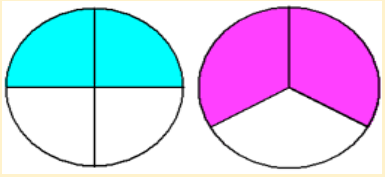
To compare and order common fractions, it is important to express them as fractions that have the same denominator. It then becomes easier to compare numerators of the fractions. If the fraction also includes a whole number, you would compare the whole number before finding the common fraction.

Note: A fraction chart is one of the most appropriate tools to use when comparing fractions.

Example:

Replace the* by <;> or = in order to show the relationship between the two fractions $\frac{2}{4}$ * $\frac{2}{3}$

Represent both fractions in the diagram and then compare them. The diagrams clearly indicate that two-thirds are greater than two quarters, therefore $\frac{2}{4} < \frac{2}{3}$



1. Replace the * with < or > or = in the following:

(b) $\frac{1}{3}$ * $\frac{1}{3}$

(c) $\frac{1}{3}$ * $\frac{1}{3}$

(d) $\frac{9}{10}$ * $\frac{3}{5}$

(e) $\frac{9}{10}$ * $\frac{4}{5}$

(f) $\frac{3}{4}$ * $\frac{10}{12}$

(g) $\frac{5}{9}$ * $\frac{1}{4}$

3. Rewrite the following fractions so that they all have the same denominator and then arrange the fractions in an ascending order.

and then arrange the fractions in an ascending order.

(a) $\frac{40}{1000}$; $\frac{3}{10}$; $\frac{60}{1000}$; $\frac{3}{10}$	
(b) $\frac{5}{6}$; $\frac{2}{3}$; $\frac{4}{9}$	
(c) $\frac{1}{2}$; $\frac{1}{4}$; $\frac{3}{5}$	
(d) $\frac{4}{5}$; $\frac{1}{6}$; $\frac{1}{3}$; $\frac{7}{10}$; $\frac{19}{30}$	
(e) $\frac{8}{9}$; $\frac{11}{12}$; $\frac{5}{6}$; $\frac{2}{3}$; $\frac{7}{8}$	

5.2 Adding and subtracting fractions and calculating fractions of a whole

Addition and subtraction of fractions

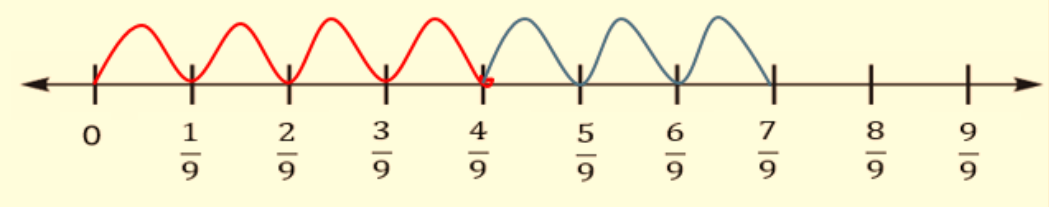
The process for adding and subtracting fractions with the same denominators is quite simple because we just need to work with the numerators. It can be done using a number line.

Example:

Calculate $\frac{4}{9} + \frac{3}{9}$

The denominator shows into how many parts one whole must be divided. Represent this using a number line. In this case, there must be 9 parts.

FINAL DRAFT COPY



Four ninths are represented by the red line and 3 ninths are represented by the blue line. The total is seven ninths.

Therefore $\frac{4}{9} + \frac{3}{9} = \frac{7}{9}$ or 4 ninths + 3 ninths = 7 ninths.

1. Calculate the following. Simplify your answer where possible.

(a) $\frac{3}{7} + \frac{2}{7}$	(c) $\frac{5}{8} - \frac{3}{8}$
(b) $\frac{5}{16} + \frac{6}{16}$	(d) $\frac{74}{100} - \frac{23}{100}$

Sometimes when you add common fractions, the result is more than a whole. The answer becomes a whole number and a common fraction and we call that a **mixed number**.

Example:

$\frac{5}{6} + \frac{3}{6}$

$= \frac{5}{6} + (\frac{1}{6} + \frac{2}{6})$ [$\frac{3}{6}$ is broken up into $\frac{1}{6} + \frac{2}{6}$

$= (\frac{5}{6} + \frac{1}{6}) + \frac{2}{6}$ [Associative property]

$= \frac{6}{6} + \frac{2}{6}$

$= 1 + \frac{2}{6}$

$= 1\ \frac{2}{6}$

2. Calculate

   (a) $\frac{9}{10} + \frac{2}{10}$

   (b) $2\frac{1}{4} + 5\frac{2}{4}$

   (c) $3\frac{1}{3} + 2$

   (d) $4\frac{8}{10} - 2\frac{7}{10}$

3. A team of road-builders built $\frac{8}{12}$ km of road in one week, and $\frac{10}{12}$ km in the next week. What is the total length of road that they built in the two weeks?

4. Margaret has $5\frac{5}{8}$ bottles of cooking oil. She gives $3\frac{3}{8}$ bottles to her friend Nelly. How much oil does Margaret have left?

5. Sipho cycled $3\frac{1}{4}$ km on his bike on Monday, $5\frac{2}{4}$ km on Wednesday and $4\frac{1}{4}$ km on Friday.

   (a) How many kilometers has he covered altogether?

   (b) If he needs to complete 20 km in a week, how many more kilometers does he need to ride?

5.3 Fraction of a whole

Calculating a fraction of a whole number is the same as multiplying the fraction by the whole number.

Example:

How much is $\frac{4}{10}$ of R70?

- Instead of writing $\frac{4}{10}$ of R70, we may write $\frac{4}{10} \times$ R70
- $\frac{1}{10}$ of $70 = 7$, therefore $\frac{4}{10}$ of 70 is the same as ($\frac{1}{10}$ of 70) $\times 4 = 7 \times 4 = 28$

Calculate the following:

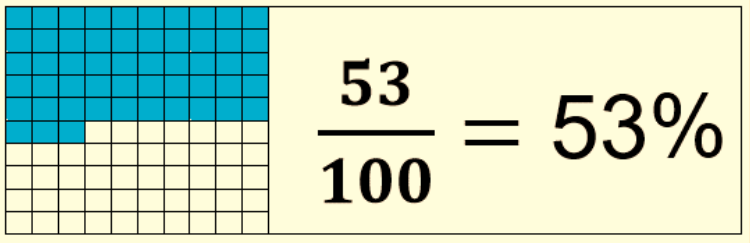
1. What is $\frac{1}{3}$ of 12 sweets?
2. How much is $\frac{1}{100}$ of R400?
3. How much is $\frac{7}{100}$ of R250? (c)
4. How much is $\frac{25}{100}$ of R600?
5. A friend bakes 12 scones and he gave Bob 4 scones. What fraction of scones did Bob eat?
6. There are 800 learners in a school. $\frac{3}{5}$ of them are boys.
   (a) How many learners are boys?
   (b) How many learners are girls?
7. $\frac{4}{4}$ of _____ balls = 20 balls.
8. What is 55% of R600

5.4 Equivalence, comparing and ordering fractions

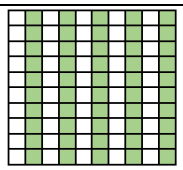
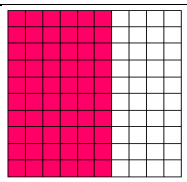
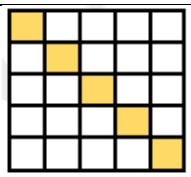
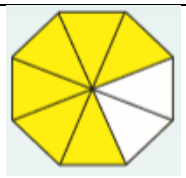
It is very easy to understand fractions expressed as **percentages**. Expressing a fraction as a percentage means the same as **converting** a fraction to a percentage. **Percent** means per hundred or **out of 100**. The percentage symbol **%** is a shorthand representation of this.

Example:

What is the meaning of 53 out of 100?. It can be expressed in a number of ways as shown:



1. Complete the following table:

Diagrammatic form	Common fraction	Percentage
		
	$\frac{3}{5}$	
		
		

2. Write each of the following percentages as a fraction. Simplify where possible.

(a) 12%

(b) 44%

(c) 63%

(d) 75%

(e) 125%

3. Express each of the following as a percentage.

(a) $\frac{12}{100}$

(b) $\frac{12}{25}$

(c) $\frac{75}{500}$

(d) $\frac{600}{1000}$

FINAL DRAFT COPY

4. Calculate:

(a) 20% of 25

(b) 60% of R15

(c) 24% of R72

(d) 20% of R300

(e) 130% of R50

(f) What percentage is 15 out of 25?

(g) If the cost price of an item is R900, 00, and the profit was 30%. What is the selling price?

(h) A farmer makes 20% profit of R6 250 when she sells 25 of her sheep. If the profit is 20 %, how much is each sheep?

5.5 Percentages including percentage increase and percentage decrease

Percentage increase or decrease is the difference between the final value and the initial value, expressed in the form of a percentage.

Examples

Percentage increase	**Percentage increase**
Calculate percentage increase if the price of a train fare increases from R50 to R65. • We first need to calculate the amount of the increase R65 – R50 = R15 • The amount of the increase is R15 and the original price is R50. Therefore, the fraction of the increase is $\frac{15}{50}$.	**Calculate percentage decrease if the price of a dress decreases from R240 to R180 during a shop sale**. • We first need to calculate the amount of the decrease R240 – R180 = R60 • The amount of the decrease is R60 and the original price is R240. Therefore, the fraction of the decrease is $\frac{60}{240}$.

FINAL DRAFT COPY

• To work out percentage increase, one needs to multiply the fraction by 100 • Therefore, percentage increase = $\frac{15}{50} \times 100\% = 30\%$	• To work out percentage decrease, one needs to multiply the fraction by 100 Therefore, percentage increase $= \frac{60}{240} \times 100\% = 25\%$

Calculate the following:

1. Items bought for R150 are sold at a loss for a price of R90. Calculate the percentage loss.
2. Joshua buys pencils for R3,30 and sells then for R5,50 each. What percentage profit does he make?
3. The production maize increases from 25 tons to 30 tons. The percentage increase in production of maize is ___.
4. Find the percentage change of the first quantity with reference to the second quantity: 72 units, 60 units.
5. Due to an error, Michael misread a number 142, as 124. What is the error in reading the number?
6. Tracksuits at a sports store are on sale for R476. If the original price was R560, what discount rate does this represent?

5.6 Calculating amounts if given percentage increase or decrease

Work out the difference (increase or decrease) between the two numbers you are comparing. Divide the increase or decrease by the original number and multiply the answer by 100.

Example: Increase R700 by 35%

The amount after percentage increase is $R700 + R700 \times \frac{35}{100}$

$= R700 + R245 = R945$

OR

The amount after percentage increase is $= R700 + (100\% + 35\%)$ of R700

$= 135\% \times R700 = R945$

FINAL DRAFT COPY

1. Answer the following:
   (a) Increase R250 by 25%
   (b) Increase R64 by 30%
   (c) Decrease 200 by 50%
   (d) Decrease 900 by 33%

2. Solve the following problems:

(a) Thabo buys and sells t-shirts. He buys the shirts at a cost of R50 each. He sells them at a profit of 75%. What is his selling price?

(b) Colleen estimates that she spends R10 to prepare a hotdog. If she wants to sell her hotdogs at 50% profit, for how much must she sell them?

(c) A wholesaler decides to give a 30% potatoes discount on all food items. A bag of potatoes originally sold for R20. For how much will they sell it?

FINAL DRAFT COPY

UNIT 6

DECIMAL FRACTIONS

8,235 can be written as 8,235 = 8 + 0,2 + 0,03 + 0,005 in expanded notation. The place values for these parts are as follows:

- the 8 represents the **units**
- the 2 represents the **tenths**
- the 3 represents the **hundredths**
- the 5 represents the **thousandths**

For example, the **value** of the 2 is two tenths and the **place value** of the 2 is **tenths**, because it represents **tenths.** A decimal can be written as a common fraction and or a percentage, for example $0{,}69 = \frac{69}{100} = 69\%$

6.1 Counting, ordering and comparing decimal fractions to at least two decimal places

1. Count forwards and backwards
   (a) Count forwards in decimals in intervals of 0,4 from 0,4 to 2
   (b) Count forwards in decimals in intervals of 0,05 from 0,27 to 0,52
   (c) Count forwards in decimals in intervals of 0,2 from 0,37 to 1,37
   (d) Count backwards in decimals in intervals of 0,07 from 0,82 to 0,47
   (e) Count backwards in decimals in intervals of 0,55 from 0,79 to -1,96

2. Fill in $<;\ >\ or\ =$
   (a) 22,1 $*$ 22,1
   (b) 12,34 $*$ 2,34
   (c) 0,10 $*$ 0,01
   (d) 5,70 $*$ 5,07
   (e) 3,6 $*$ 3,62

3. Arrange the following temperatures from coolest to hottest.

Day	M	T	W	T	F	S
Temperature $°C$	14,8	23,6	13,2	-2	18,32	$-4{,}33$

4. Arrange the following decimal fractions in descending order.
   (a) 2,43;1,76,1,61;1,04;1,53;1,21;1,4
   (b) 0,21;0,06;0,6;0,16;0,08;0,8;0,41

FINAL DRAFT COPY

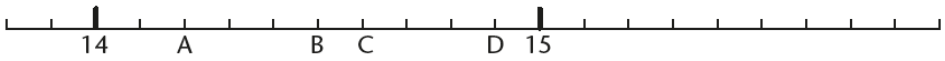
(c) 1,18;1,06;0,7; 2,057;1,02;1,2;1,61

(d) 23,7;23,07;23,5;23,90;23,06

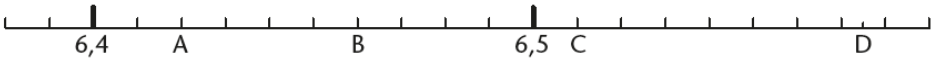
(e) 6,49;6,93;6,09;6,6;5,88,5,8

5. Write the value of the letters A, B, C,D using a number line

(a)



(b)



6.2 Calculations with decimal fractions:

Addition and subtraction of decimals

1. Add or subtract decimals

(a) 23, 45 + 12,5 + 1,677

(b) 17,12 + 0,23 + 2,01

(c) −31,59 − 64,21

(d) 214,06 − 10,01

(e) 498,34 − 762,02

2. Calculate.

(a) 2,15+8,21 – 7,21

(b) 9,43 – (3,61 + 1,14)

(c) 1, 75 m – 1,2 m + 0, 256 m

(d) 359,88 − 12,27 + 221,6

(e) 28 cm + 305,08 cm − 111,052 mm

3. Multiply

(a) 4,8 × 10

(b) 12,21 × 10

(c) 144,36 × 100

(d) 9,49 × 100

(e) 1,89 × 10

FINAL DRAFT COPY

6.3 Recognize equivalence between common fraction, decimal fraction and percentage forms of the same number

1. Complete the table

Common fraction	Decimal fraction	Percentages
$\frac{1}{2}$		50%
	0,8	80%
	0,1	
$\frac{3}{4}$		
	0,33	33%
$\frac{3}{8}$		37,5%

6.4 Consolidation activity

Solve the following:

1. A roll of material is 25 metres long. 1,35 metres of material is needed to make one dress.
   (a) How many dresses can be made from the roll of material?
   (b) How much material will be left over?
2. A farmer owns a land with an area of 25,05 hectors and is given a further 30,07 hectors of land. He then inherits 3 more hectors from his father. What is the total area of his land?
3. Zamokuhle is an athlete. His most recent records for 100 m sprints are: 10,54 seconds; 9,33 seconds and 11,15 seconds respectively.
   (a) What is the difference between Zamokuhle shortest record and longest record?
   (b) What is the sum of his most recent records?

UNIT 7

WHOLE NUMBERS: FINANCE

FINAL DRAFT COPY

7.1 Solving problems in financial context that involve profit, loss and discount

1. Copy and complete the table

(a)	Item	Original price	% profit	Selling price
	Cakes	R80,00		R120,00
	Packet of sweets	R35,00	25 %	
	Packet of birthday balloons		50 %	R60,00

(b)	Item	Original price	% loss	Selling price
	A bunch of bananas	R15,00		R12,00
	A plate of samp, meat and vegetables		25 %	R45,00
	A box of sausage rolls	R150,00	40 %	

(c)	Item	Cost price	% discount	Sale price
	Laptop	R6 990	35 %	
	Cellphone	R1 300		R900
	Printer		25 %	R1 500

2. Solve the word problems

(a) Nelly bought a packet of ginger juice powder for R18. The powder makes 30 litres of ginger juice. She sells the juice at a fundraising event at church for five rands a litre and hired a stall for R50. What is her percentage profit?

(b) Pinky wants a space to rent for her laundry business. She got a house for R4 000 per month. The landlord offered a 30 % discount, if she can pay cash for the first three months. How much must she pay in order to get the 30 % discount?

(c) Vuyo bought a delivery bakkie for R125 000.After driving it for three months, it gave him problems. He decided to take it back to the dealership and ask for the refund as it was still under warranty. After negotiating with the dealership, they refunded him R80 000. Calculate the percentage loss.

7.2 Solving problems in financial context that involve budgets

FINAL DRAFT COPY

1. Mandla wants to develop his barber shop business. He asked the District Youth Development Agency to fund him. They requested him to provide his budget so that they can assist him. He does not have a budget but he knows the money he receives per month and how he spent it.
   (a) Complete Mandla's budget using the following information:
   (b) Taxi fare to the barber shop per day is thirty rands per day.
   (c) Refreshments is eight hundred rands per month
   (d) He receives three hundred and fifty rands social grant.
   (e) He offers fifty rands each Sunday to his church.
   (f) He receives a four hundred rands allowance from his parents.
   (g) He spends one thousand and two hundred rands a month on toiletries.
   (h) The monthly rent for his barber shop is one thousand and five hundred rands.
   (i) The monthly maintenance for his machines is seven hundred rands.
   (j) His weekly income is one thousand eight hundred and fifty rands.

INCOME		**EXPENDITURE**	
Item	**Amount**	**Item**	**Amount**

1. Mr Vanda's grass cutting business is making **nineteen thousand two hundred rands** a month. Weekly he spends **five hundred rands** on petrol**, eighty rands** on two-stroke oil, and five hundred rands for his helper. Monthly he spends **seven hundred** rands for the string and **eight thousand** for his family's needs. Compile his monthly budget and decide if the **nineteen thousand two hundred rands** will be sufficient for his monthly expenses.

7.3 Solving problems in financial context that involve accounts and loans

Solve the following problems.

FINAL DRAFT COPY

1. Elise borrowed R25,000 from a cash loan for a period of eighteen months, at a simple interest rate of 12 % . How much must she repay at the end of the term.
2. Study Mr Fowler's cellphone account statement and answer questions

Statement

Inquiries:
Tel: 021 344 7453
Email: cellphone@acc.net

Mr D Fowler
P O Box 13426
Cape Town
8000

Email address	dfowler@gmail.com
Statement date	15 - 07 - 2022
Billing account	BA 222769
Previous account	PA 222768
Primary contact	021-3432121

Statement

Statement includes all transactions from 14 - 06 - 2022 to 14 - 07 - 2022

Previous balance		Adjustments (credit\debit)		Charges this month		Payments received (thank you)		Amount due (VAT incl.)
R2,068.31	+	R0.00	+	R1,031.18	-	-R1,039.01	=	R2,060.48

Account aging Total overdue : R 0.00

120 days overdue	90 days overdue	60 days overdue	30 days overdue	Current due	Total due (incl. overdue)
R0.00	R0.00	R0.00	R0.00	R1,031.18	R2,060.48

Statement details below are VAT inclusive

Date	Transaction	Reference	Amount	Due Date	Open Amount
14-06-2022	Invoice	I500000162876413	1,031.18	01-02-2022	1,031.18
14-07-2022	Past Invoice	I500000149977609	1,029.30	01-01-2022	1,029.30
01-08-2022	PAYMENT	P5000000180732934	-1,039.01		

Total amount due: **R2 060, 48**

a) What is Mr Fowler's cellphone number?
b) What is his previous balance?
c) What is the billing period for this statement?
d) How much is due on the first of August?
e) Does the amount due include or excludes VAT?

FINAL DRAFT COPY

TERM 2

FINAL DRAFT COPY

UNIT 1

WHOLE NUMBERS: COUNTING, ORDERING, COMPARING, REPRESENTING AND PLACE VALUE

1.1 Counting, Ordering, Comparing and Representing

Counting

1. Complete the following by counting forwards and backwards

   (a) 22 132 570; 22 133 815; 22 135 060; _____

   (b) 8 345 271; 8 345 304; _____; 8 345 370

   (c) 32 746 188; 32 746 165; _____; 32 742 119

Ordering

2. Write the numbers in ascending order.

   a) 331 136; 329 245; 21 645; 360 001; 331 111;392 245

   b) 5 561 344; 5 156 344; 5 561 434; 5 516 344; 5 561 443

Comparing

3. Replace the * with $>$; $<$ *or* $=$ to make the mathematical sentence true.

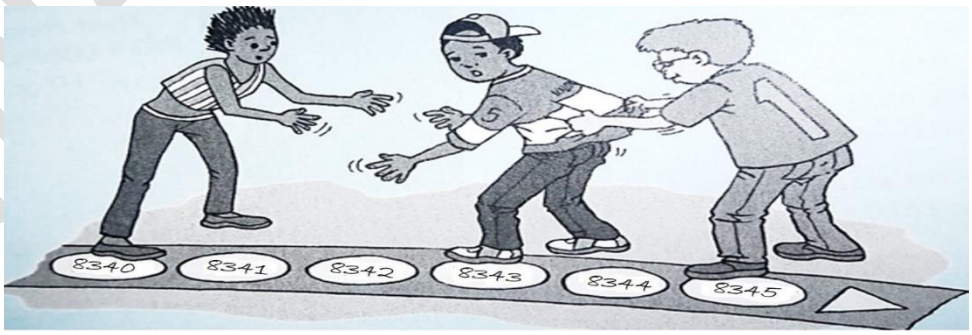
   (a) 91 457 132 * 91 457 213

   (b) 9 111 161 * 1 101 615

   (c) 132 570 * 132 570

1.2 Rounding-off

1. Study the picture and indicate if the boy in the middle should move backwards or forwards if he rounds off his position to the nearest 5



FINAL DRAFT COPY

2. Copy and complete the table.

Round off to the nearest	5	10 000	100 000	1 000 000
47 526				
235 079				
378 900 352				
976 315 463				
13 415 978				

1.3 Place value

1. Write the place values for each of the underlined digits:
   (a) 9**7** 832 435
   (b) **4**90 841
   (c) 6 4**8**7 032
   (d) 11 673 8**8**0
   (e) 88 091 **0**23

2. Write the value of each underlined digit:
   (a) **4**4 567
   (b) 6**5** 567 567
   (c) **1**1 757 458
   (d) 89 **5**61 337
   (e) 70 70**2** 670

1.4 Problem Solving

Solve the following problem.

1. Peter has four digits and they are 5, 2, 7, 6. Make the highest and smallest four digit numbers he can make without repeating the numbers.
2. Bontle bought shoes for R249 and a dress for R399. Round off each amount to the nearest ten rand.

FINAL DRAFT COPY

3. I am a six-digit number, the sum of my ten thousand, hundred thousand and thousands digits is an odd number which is a square root of one hundred and twenty-one and a factor of fifty-five. The difference between my tens and my unit digits is two. Who am I?
4. Given the number 335 436.
   (a) Find the product of the tens digit and the hundreds digit
   (b) Add the difference between ten thousand digit and hundred thousand digit to the product in (a).
5. You have a five-digit number, after rounding it off to the nearest 1000, the answer is a six-digit number. What was the possible original number?
6. The cyclists travelled 2 745 m on the first day, another 679 456cm on the second day and 345 445cm on the third day.
   (a) On which day did they cycle the furthest?
   (b) On which day did they cycle the least?
   (c) Round off the total kilometers cycled to the nearest thousandths

FINAL DRAFT COPY

UNIT 2

WHOLE NUMBERS: FINANCE

2.1 Solve problems that involve whole numbers, percentages and decimal fractions in financial contexts

Value Added Tax (VAT)

The acronym VAT stands for Value Added Tax. It is a tax levied on goods and services at each stage of production and distribution of sale from the primary to the secondary consumer. A value added tax, known in some countries as goods and services tax, is a type of tax that is assessed incrementally. It differs from country to country, in South Africa, the VAT standard rate is currently 15%. In South Africa some goods are zero rated.

What is zero-rated VAT South Africa?

Zero rated goods are products that are exempt from value added taxation (VAT).

Examples of VAT – exempt goods (zero rated goods)

Dried fruit, samp, maize meal, rice, brown bread, rated goods, edible legumes, mealie rice, pilchards in tins, eggs, milk, dried mealies, dairy powdered milk, dried and powdered milk, brown wheaten meat, bread flour and sanitary pads

Who must pay VAT in South Africa?

All consumers are legible to pay VAT directly or indirectly. When a company sells goods or services, they generally charge VAT to their customers as part of their selling price**.** It is mandatory for **any business** to register for VAT if the income earned in any consecutive twelve-month period exceeded or is likely to exceed R1 million. Any business may choose to register voluntarily if the income earned, in the past twelve-month period, exceeded R50 000

1. How much will these items be including VAT? Some of them are zero rated.

No.	Items	Price excluding VAT	Price including VAT
(a)	5 ℓ cooking oil	R 256,90	
(b)	500 g margarine	R 26,50	

No.	Items	Price excluding VAT	
(c)	500 $m\ell$ liquid soap	R 24,99	
(d)	75 $m\ell$ tooth paste	R35,95	
(e)	5 kg rice	R89,99	
(f)	1 kg sausage	R 99,95	
(g)	2 ℓ milk	R37, 49	
(h)	500 g yoghurt	R18,99	

(a) Monique buys a hairdryer for R395 excluding VAT. How much will she pay in total including VAT?

(b) Sizwe bought an iPhone 11 for R13 000 excluding VAT.He pays cash and gets a 5% cash discount. How much will he pay in total including VAT

2.2 Simple interest, hire purchase and exchange rate

Interest

In finance, interest is the money charged for borrowing money. It is expressed as a percentage of the borrowed amount. There are different types of interest e.g. simple interest and compound interest.

If you have some money, you could either keep it in your piggy bank account with no interests. If you deposit it into a bank account, you are effectively lending money to the bank. Because you are lending the bank money, you expect some extra money back. This is known as interest. Similarly, if you borrow money from a bank, then you can expect to pay interest on loan. Interest is charge at a percentage of the money owed over the period of time it takes to pay back the loan. This means that the longer the loan exists; the more interest will have to be paid on.

What is simple interest?

Simple interest is the interest calculated only on the initial amount that you invested over a certain period. When someone lends money to someone else, the borrower usually pays a fee to the lender. This fee is called 'interest'. Simple or flat rate interest is usually paid each year as a fixed percentage of the amount borrowed or lent at the start.

FINAL DRAFT COPY

The simple interest formula is as follows:

Simple Interest = Principal (P) amount × Rate (R) × Time (T), where:

- Simple Interest: is the total amount of interest to be paid
- Principal (P): is the amount lent or borrowed
- Rate (R): is the percentage of the principal charged as interest each year
- Time (T): is the period in years to pay back the loan.

What is Hire Purchase?

Higher purchase is a system buyer where a buyer pays for an asset in regular instalments while enjoying the use of it. The buyer has an option to make an initial down payment and pays the balance plus interests in instalments. During the repayment period ownership of the item does not pass to the buyer (it is on 'hire'). Upon the full payment of the loan plus interests the title passes to the buyer (the purchase is now complete).

(Gr 8 DBE workbook 1 Pg 20)

The repayment period is normally the same as the production life of the item bought on hire purchase. For example, a farmer buys a tractor and pays it off over five years. After five years he typically has to replace the tractor.

Solve the following problems.

1. Ali borrowed R 5 000 from a bank over a period of 3 years at a simple annual interest rate of 8,5%.How much must he repay at the end of the term period
2. Bulela wants to buy a racing bicycle for R 2 400 cash. He is having R1 800. What annual rate return will he need to invest to accomplish his goal in two years?
3. Karin, a farmer wants to buy a new tractor. The tractor costs R340 000 including VAT. She can pay a deposit of R50 000. She decides to buy the tractor on hire purchase over 60 months at a simple interest rate of 10%.
   (a) How much interest will she pay?
   (b) How much will she pay in total for the tractor over 60 months?

(c) What will her instalment be?

Exchange Rate

An **exchange rate** is the current market price for which one currency can be changed for another. Currency is money in the form of paper or coins usually issued by a government and generally accepted at its face value and as a method of payment. Different countries have their own currencies.

A currency gets stronger when the money is invested in the country. When we buy products that are made in South Africa, we are investing in South African business and keeping the money in the country. When we buy products imported from other countries, we are investing money in those countries and as a result, the rand will weaken. The more South African products we buy, the greater the demand for them will be and more jobs will become available for South Africans. The price of oil increases when the South African rand weakens. This is because when the rand is weaker, we can buy less of other currencies with the same amount of money.

Exchange rates are determined by factors such as, economic growth, interest rates, confidence, inflation rates, etc. The central bank is mandated by the government to determine exchange rates. The rate is set against another major world currency.

Country	Currency	Sign	Code
South Africa	Rand	R	ZAR
United States of America	United States Dollar	$	USD / US$
European Union	Euro	€	EUR
United Kingdom	Pound	£	GBP
Canada	Canadian Dollar	$	CAD
Australia	Australian Dollar	$	AUD / A$

Solve the following problems.

4. Rita lives in Umlazi in South Africa, her brother lives in London in England. His brother deposited her £70 for her birthday. How many South African Rands (ZAR) can she buy with her birthday money? (Rate £1 = R20,26).

5. Collen earns R800 from his part – time job. He wants to buy a t- shirt which costs $30 USD. After his purchase how much money will he have left in ZAR? (Rate $1 = R17,09).

6. Lulamile wants to finish her engineering course in Germany, the total amount needed for the course is R150 000. BMW gave her a sponsor of €8 000. How much in rands does Lulamile needs to raise now? (Rate €1 = R17,22).

FINAL DRAFT COPY

UNIT 3

ALGEBRAIC LANGUAGE

An **algebraic expression** is made up of numbers, variables and operation signs. Algebraic expressions can also be described in words, flow diagrams and tables.

Equivalent expressions are algebraic expressions that have different sequences of operations, but have the same numerical value for any given value of x.

A convention is something that people have agreed to do in the same way. We write $3 \times x$ rather than $x \times 3$, and we write $3x$ rather than $x3$. In algebraic expressions, when multiplying with variables the multiplication sign is often not shown: We write $4x$ instead of $4 \times x$ and $4(x - 5)$ instead of $4 \times (4(x - 5)$

Possible misconceptions:

- $x + x = 2(x$ and **NOT** x^2 (note the convention is to write $2x$ rather than $x2$)
- $x^2 + x^2 = 2x^2$ and **NOT** $2x^4$

1. Translate the following sentences into algebraic language
   (a) The sum of a number and eight is multiplied by six.
   (b) The product of a number and 3 is increased by 22.
   (c) Add the sum of a and b to the product of c and d.
   (d) 7 multiplied by h.
   (e) The product of a number and 34 added to 12.
   (f) The difference between two hundred and 137.
   (g) Divide a number by 2 and subtract 9 from the answer.
   (h) One third of b.
   (i) Add 2 to the product of double a number and seven.
   (j) The sum of a number squared and its square root.
   (k) Seven less than the cube root of a number.
   (l) The difference between the square of a number and three times the cube root of the number.
   (m) What is different between $3(x + 5)$ and $3x + 5$.

FINAL DRAFT

2. Interpret and write the following rules (algebraic expression) in words.

(a) $\frac{1}{2}\,n\,+2$

(b) $10(n\,+3)$

(c) $4n-5$

(d) $(l\times 3)+22$

(e) $w^2+\sqrt{w}$

3. Complete the table

No.	Expression	Variable	Constant	Coefficient of
(a)	$4x^2+6\,x-10$	x	-10	
(b)	$6s^3+s^2+5$			
(c)	$4p^7$			
(d)	$\frac{y^2+1}{2}$			
(e)	$\frac{k}{3}-15$			

4. Consider the expression $3x^2+4\,y-4\,xy+7y^2+22$

(a) What is the coefficient of the fourth term?

(b) What is the constant in this expression?

(c) What is the variable in the second term?

5. Rewrite the following expressions, applying algebraic conventions:

(a) $3\times(11-x)+(2\times x+3)13$ $\quad 3\times(11-x)+(2\times x+3)\,13$

(b) $-2\times a+b$

(c) $a2$

6. Solve by inspection and check each answer by substitution

(a) $v\,+10=44$

(b) $n-4=6$

(c) $7t+8=9$

(d) $\frac{x}{5}+9=12$

(e) $x^2=16$

FINAL DRAFT COPY

(f) $a^3 = 125$

7. Solve the following equations using trial and improvement

(a) $x - 6 = 10$

x	$x - 6$	Comment

(b) $x^3 - x = 120$

(c) $4a - 3a + 5 = 12$

8. Substitute to determine the numerical value of the following expressions.

No.	**Expressions**	**Substitute by:**				
(a)	x	**−2**	**−1**	**0**	**1**	**2**
(b)	$3x - 2$					
(c)	$3x + 2 + 2x - 3$					
(d)	$(3x + 2)(2x - 3)$					
(e)	$\frac{(3x + 2)(2x - 3)}{3x + 2}$					
(f)	$\frac{6x^2 - 5x - 6}{3x + 2}$					

9. Solve the following problems.

(a) Lebo is a florist. She makes a certain number of bouquets, and sells them at R40 each. The money received from the sale of bouquets is R2 000. How many bouquets did Lebo make?

FINAL DRAFT COPY

(b) A builder delivers a number of tons of timber to a customer. He has a large truck that holds three tons of timber and a small truck that holds two tons of timber. The driver makes six journeys with the large truck and one journey with the small truck. Each time the truck is fully loaded. How many tons of timber did the builder deliver to his customer?

(c) Naledi sells six cups of hot chocolate for R30. She writes the record of her sales in the equation $6x = 30$. If x is the cost of each cup of hot chocolate, at what price will Naledi sell each cup?

(d) Mazwi and Themba pick pears off the tree. For every 17 pears that Mazwi picks, Themba picks 20. Together they picked 148 pears.

(i) Represent that in an algebraic equation

(ii) Calculate how many pears did Themba pick?

FINAL DRAFT COPY

UNIT 4

NUMERIC AND GEOMETRIC PATTERNS

4.1 Geometric patterns

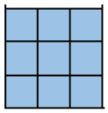
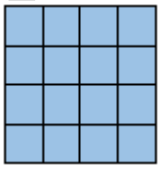
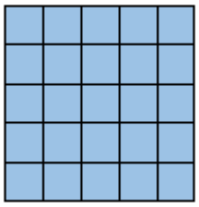
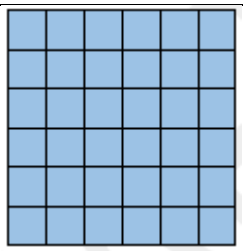
The **"structure"** in geometric patterns refers to the physical arrangement of geometric shapes at every stage of the pattern.

The **structural analysis** is very useful to "see" the **recursive relationship** to help formulate the general rule for the pattern even before the table or flow diagram can be drawn. A calculation plan can be derived from the structure and subsequently the rule.

1. Study the figures of tiles below, draw tile number 4 and describe tile no.5

Tile no. 1 *Tile no. 2* *Tile no. 3* *Tile no. 4*

2. Write the calculation plan and the general rule for each tile. The first one has been done.

	Tile **1**	Tile **2**	Tile **3**	Tile **4**
Structure				
Calculation plan	$3 \times 3 = 9$ $(\mathbf{1} + 2)^2 = 9$			
Rule				

3. Study the diagrams

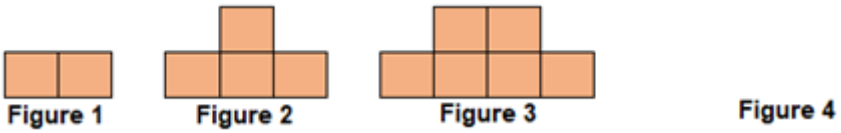
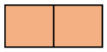
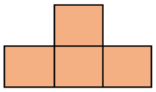
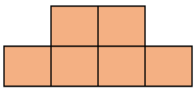
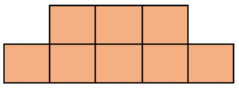


Figure 1 Figure 2 Figure 3 Figure 4

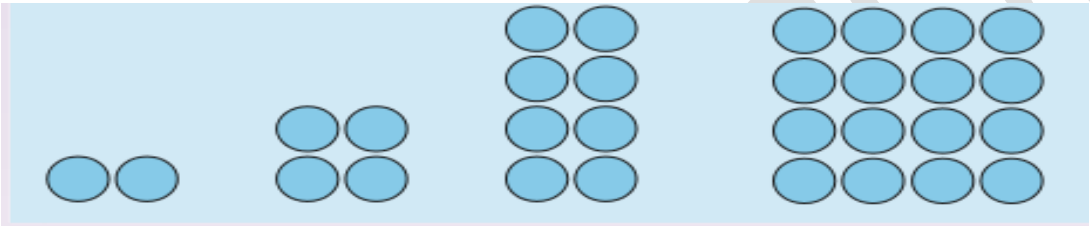
(a) Extend the pattern by drawing figure 4

(b) Write the calculation plan

(c) Describe the pattern in words.

	Figure 1	Figure 2	Figure 3	Figure 3
Structure				
Calculation plan				
Rule				

4. Study the pattern below



(a) Represent the pattern in table

Position	1	2	3	4
No. of balls				

(b) Write the general formula or rule

(c) Describe the pattern in words.

5. Investigate how the pattern grows and answer the questions below

Picture1	Picture 2	Picture 3	Picture 4	Picture 5

(a) Extend the pattern by drawing figure 5

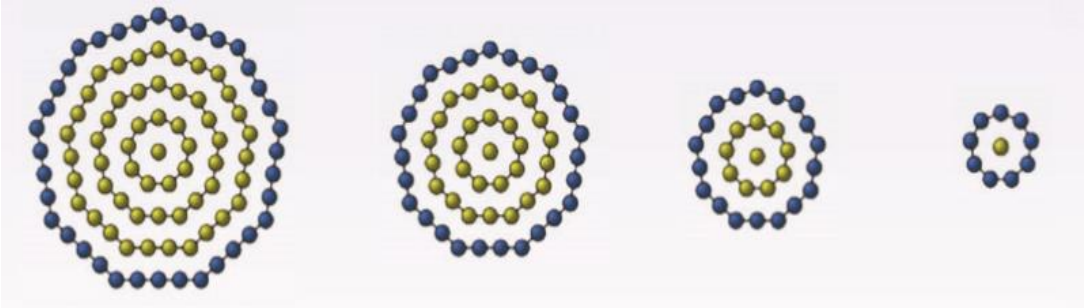
(b) Describe the pattern in words

(c) Represent the pattern in a flow diagram

(d) Write the general rule

FINAL DRAFT COPY

6. Study the pattern below and answer the questions that follows:



(a) Represent the pattern in a table

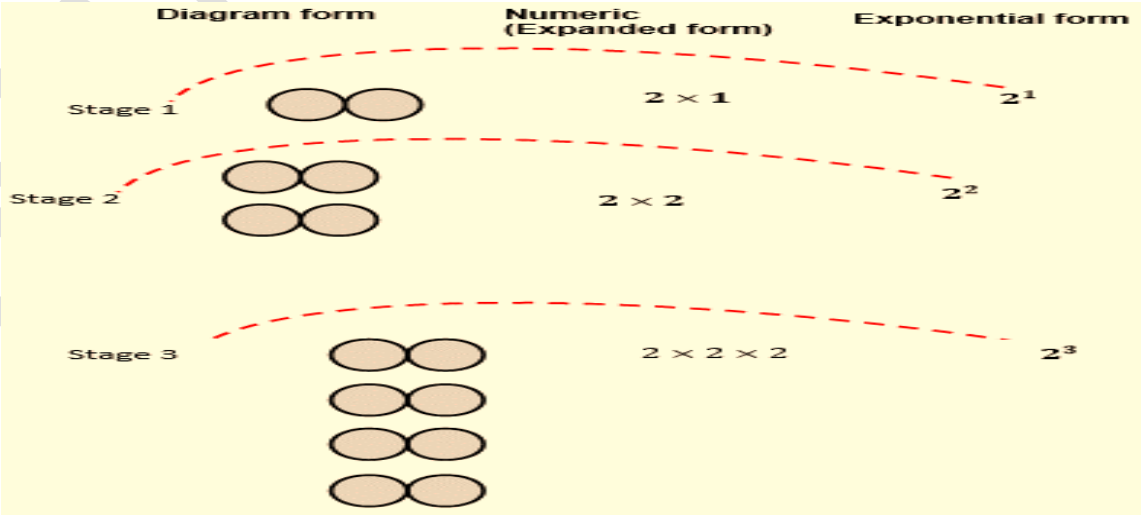
Position				
No. of beads				

(b) How is the pattern in question 6 different from the pattern in question 5?

4.2 Numeric patterns

There is a connection between the **physical structure** (construction) and the **numeric representation** at each stage of the pattern. Studying and analyzing the structure assists in developing a numeric pattern and subsequently the general rule.

Example:



Numeric representation in a table.

Stage number	1	2	3	4	5	n
Number of circles	2	4	8	16		
Equivalent description	2 × 1	2 × 2	2 × 2 × 2	2 × 2 × 2 × 2		
Equivalent description	2^1	2^2	2^3	2^4	2^5	2^n

1. What are the next three numbers in each of these patterns?

   (a) 15; 20; 25; 30; _____; _____: _____

   (b) 8; 16; 24; 32; _____; _____; _____

   (c) 3; 15; 27; 39; _____; _____ ; _____

   (d) 0,9; 0,7; 0,5; 0,3; _____; _____; _____

   (e) $\frac{1}{4}$; $\frac{1}{2}$; $\frac{3}{4}$; _____; _____; _____

   (f) 42;35 ;28;21; _____; _____; _____

   (g) 128;116;104; 92; _____; _____; _____

Apply the knowledge learnt in number sentences and algebraic language to interpret the following:

2. Extend the following patterns and indicate the rule

(a) 3; 8; 13; 18; _____; _____

(b) 10; 14;18; 22_____; _____

(c) 10; 25; 40; 55; _____; _____

(d) 5; 10; 15; 20; _____; _____

(e) 2; 4; 8; 16 _____; _____

(f) 15; 25; 35; 45; _____; _____

(g) 1;4;9;16; _____; _____;

3. In what way are patterns B and E different?
4. Given that any term in a number pattern is : $Tn = 2 \times n + 1\ Tn$

   (a) Determine the value of the 1st term

   (b) Determine the value of the 4th term

   (c) What will be the value of the 100th term?

FINAL DRAFT COPY

(d) Write the rule for this pattern in words.

5. Which of the following rules will produce the number pattern given in the second row of the table below?

   A. Term value = $8n$ where n is the term number

   B. Term value = $6n - 1$ where n is the term number

   C. Term value = $6n + 2$ where n is the term number

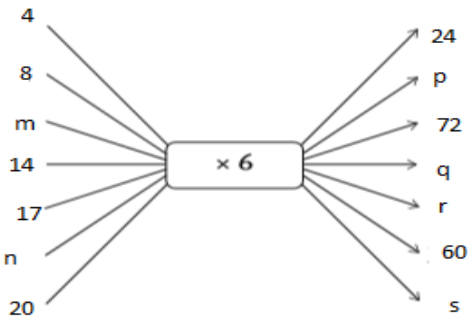
   D. Term value = $10n - 2$ where n is the term number

   E. Term value = $5n + 3$ where n is the term number

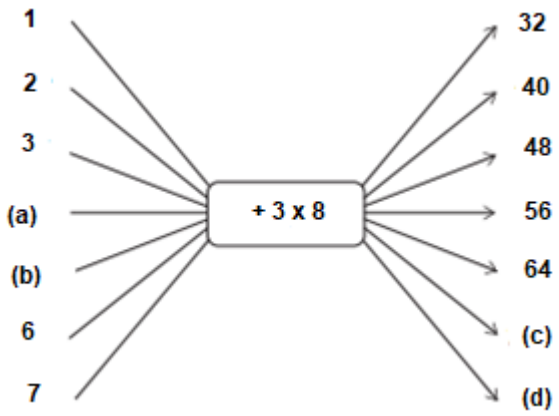
Term number	1	2	3	4	5	6
Term value	8	13	18	23	28	33

Input and output

6. Copy and complete the following flow diagram:



7. Fill in the missing output and input numbers:



(e) Describe the pattern in your own words.

FINAL DRAFT COPY

Problem solving

8. Mr Twala pays a fee to park his car in a parking lot every day. He has to pay R3 to enter the parking lot and then a further R2 for every hour that he leaves his car there.

   (a) Copy and complete the following table to show how much his parking costs him per day for various numbers of hours

Number of hours	1	2	3	4			7
Cost of parking in R			R9	R11	R13	R15	R17

   (b) How did you complete this table? Describe your method in words.

   (c) Is there another way that you could complete the table? Describe the rule.

9. Thapelo is building a house model using cubes. If he uses 2 cubes in the first model, 5 in the second and 10 in the third model.

(a) How many cubes will he need to complete the fourth model, if the pattern continues?

(b) Write down a mathematical rule to determine the value of any term in the sequence.

(c) Use your rule to determine the 10th term of the pattern.

10. Write the symbolic formula for the following statements:

(a) Output number is equal to input number multiplied by four, added to twelve

(b) Output number is equal to 2, multiplied by the input number added to four

(c) Create your own numeric pattern in a table and a flow diagram

(d) Write the formula for your pattern.

(e) Write your pattern in a sequence.

(f) Describe the pattern that you created in words.

FINAL DRAFT COPY

UNIT 5:

MEASUREMENT

5.1. Length

1. Convert the following lengths to the given units.
   (a) 1,2 m to cm
   (b) 13 478 mm to cm
   (c) $3\frac{1}{2}$ m to mm
   (d) 639,2 cm to mm
   (e) 4 593 cm to m
   (f) 4 071 mm to cm

2. My car has to go for a service in 2 871 km. I drove 1 264 km during the month. How many kilometres left before I have to take my car for the service?

(DBE WORBOOK 2 Grade 6, Page 97 - 98 of 11th Edition)

3. Bongi travelled 3,5 km to the event and Sipho travelled 3 250 m to the same event. Who travelled the furthest between the two? Write your answer in metres.
4. David measured the distance between two trees with a stick. He found that the distance was $12\frac{1}{3}$ sticks. When he measured the stick with his ruler, it was 78 cm long. What was the distance between the two trees? Give your answer in centimetres and metres.
5. Calculate:

(a) Calculate in centimetres and round off your answer to the nearest 100 mm:
5 m + 157 cm − 1 145 mm.

(b) Calculate in millimetres and round off your answer to the nearest 1 000 mm:
23 × (1 380 mm − 78 cm)

FINAL DRAFT COPY

(c) Calculate in metres and round off your answer to the nearest kilometre:
5,4 km − 204 m × 14.

5.2. Mass

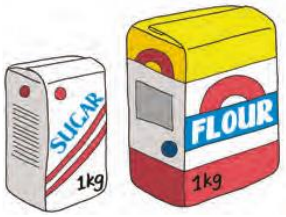
1. Calculate the following problems in your writing books:

(a) A 500 g bag of flour costs R3,50. How much will a 1,5 kg cost?

R3,50

(DBE WORBOOK 2 Grade 6, Page 97 - 98 of 11th Edition)

2. Compare a 1 kg packet of sugar and a 1 kg packet of flour.



DBE Textbook
Gr 6 LB, Page 216

(a) Does the mass of the two packets differ?

(b) Does the size of the two packets differ?

(c) Explain why it is that a 1 kg packet of sugar and a 1 kg packet of flour have different sizes.

3. Compare the mass of the packets of sugar below and answer questions that follow.

A: 1 kg B: 250 g C: 2,5 kg D: 5 kg E: 500 g

(a) How many of Packet B have the same mass as Packet A?

(b) What fraction of the sugar in Packet A has the same mass as Packet B?

(c) Which packet holds half as much sugar as Packet A?

(d) Which packet holds $2\frac{1}{2}$ times as much sugar as Packet A?

FINAL DRAFT COPY

5.3. Capacity/Volume

1. Diesoline is used to generate electricity at a small power station. The power station uses 684 ℓ of diesoline each day. For how many days can the power station operate if there is a stock of 9 765 ℓ of diesoline available?
2. Thuli adds 250 mℓ of concentrated fruit juice to 2 ℓ of water, to make drinks for the athletes in a long-distance race.
   (a) How much concentrated juice should she add to a 5 ℓ of water?
   (b) How many athletes can she provide with 400 mℓ of juice each, with the juice she made by adding concentrate to 5 ℓ of water?
3. A bucket holds 3,2 litres, a second holds 575 millilitres and a third holds 4 litres 920 millilitres. What is the three containers total capacity? Write the answer in millilitres.
4. 30 teachers from the same school organised 8 times 2,25 litres of soft drinks. If they share it equally, how much each is going to drink in millilitres?
5. Mila and Bongi are selling water in bottles, Mila sold 750 millilitres of water and Bongi sold 4 250 millilitres.
   (a) How much more did Bongi sell than Mila?
   (b) Write the answer in litres.
   (c) Write the answer as a common fraction.
6. Match the common fractions below with the mass of the illustrated items.

$\frac{1}{4}$ kg; $\frac{1}{2}$ kg; $\frac{3}{4}$ kg; $2\frac{1}{2}$ kg; $1\frac{1}{2}$ kg

7. Find the equivalent mass of the following objects (in grams):
   (a) a baby with a mass of 2,8 kg
   (b) a book with a mass of 0,5 kg
   (c) a brick with a mass of 1,5 kg

FINAL DRAFT COPY

8. Find the equivalent mass of the following objects (in kilograms):
   (a) a 250 g bag of flour
   (b) an apple of 100 g
   (c) a 750 g stone

9. A bag of 30 oranges, all of the same mass, weighs 5 kg. Calculate the mass of:
   (a) 15 oranges
   (b) 5 oranges
   (c) 1 orange.

FINAL DRAFT COPY

TERM 3

FINAL DRAFT COPY

Contents

TERM 3
Table of Content

Unit 1
1.1 Geometry of Straight lines……60
1.2 Angle relationship……64

Unit 2

Geometry of 2-D Shapes…..……..65

Unit 3

Transformation……68
3.1 Use transformation to make composite shapes and tessellations…….68
3.2 Recognise, describe and perform translations, reflections and rotations with geometric figures and shapes……69
3.3 Enlargement……70
3.4 Describing pattern……...73

Unit 4

Geometry of 3-D Objects……74
4.1 Describe, sort and compare……75
4.2 Building 3-D models using nets…..……..77

Unit 5

Areas and Perimeter……79
5.1 Calculate the perimeter of regular and irregular polygons……..79
5.2 Use appropriate formular to calculate the perimeter squares and rectangles……80
5.3 Use appropriate formulae to calculate the perimeter of triangle and circles…..81
5.4 Use appropriate formulae to calculate the area of squares and rectangles……83
5.5 Use appropriate formulae to calculate the area of triangles and circles……..84
5.6 Problem solving involving Perimeter and area of polygons……..85
Unit 6
Surface Area, Volume and Capacity……86
6.1 Using appropriate 2formulae to calculate the surface area, volume and capacity of
cubes……86
6.2 Using appropriate formulae to calculate the surface area, volume d capacity of rectangular
prisms……..86

FINAL DRAFT COPY

6.3 Using appropriate formulae to calculate the surface……..
87
6.4 Using appropriate formulae to calculate the surface area, volume and capacity of
,cylinders……88
6.5 Solving problems in context involving……..……88

Unit 7

Data handling……89

7.1 Collect and organise data……..……89
7.2 Summarise Data……..……91
7.3 Represent Data……..……93
7.4 Interpret and reporting……..……98
7.5 Report……100

FINAL DRAFT COPY

UNIT 1

1.1 GEOMETRY OF STRAIGHT LINES

1. Match the term in column A with the definition in Column B

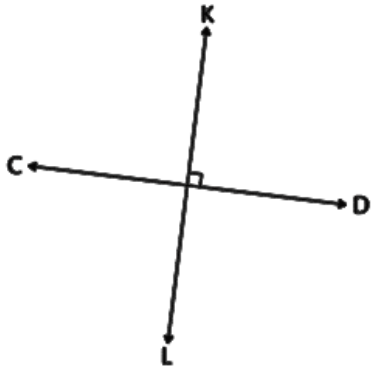
Column A	Column B

(a) Ray	i. A set of points with no definite starting-point and end-point.
(b) Straight line	ii. When two lines on the same plane are a constant / same distance apart
(c) Line segment	iii. When two lines intersect at right angle
(d) Parallel line	iv. A set of points with a definite starting-point and no definite end-point
(e) Perpendicular line	v. A set of points with a definite starting-point and an end-point.

2. Identify the types of lines below:

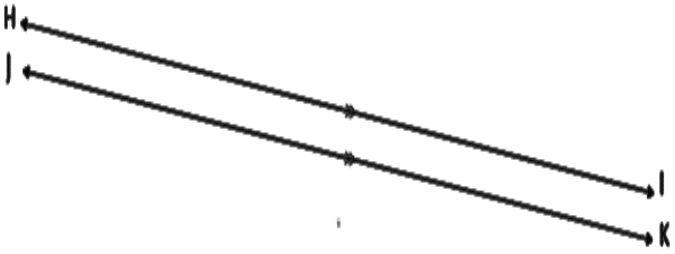
(a)		
(b)		
(c)		

(a)



Line CD is _____ to Line KL

(b)



FINAL DRAFT COPY

Line HI is _____ to line JK

3. Look at the following pictures, mark the parallel and perpendicular lines

No.	Picture	
(a)		
(b)		
(c)		

4. Identify types of lines are formed in the following pictures below?

(a)		
(b)		

1.2 ANGLE RELATIONSHIPS

FINAL DRAFT COPY

Remember:

The sum of the angles on a straight line is 180°

If lines are perpendicular, then adjacent supplementary angles are each equal to 90°.

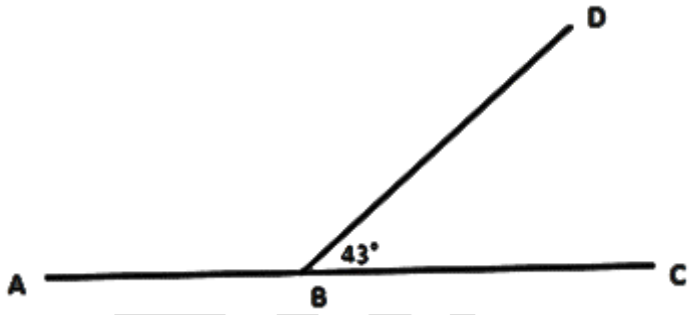
If lines intersect, then vertically opposite angles are equal.

If parallel lines are cut by a transversal, then

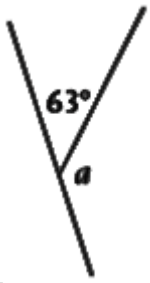
- corresponding angles are equal
- alternate angles are equal
- co-interior angles are supplementary

1. Given the diagrams below

   (a) Calculate angle ABD if angle DBC is equal to 43°

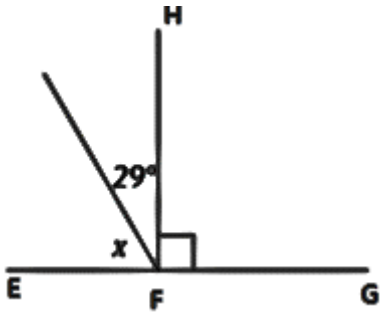


   (b) Calculate Angle a

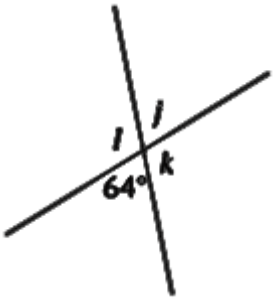


   (c) If line HF is perpendicular to line EG calculate angle x

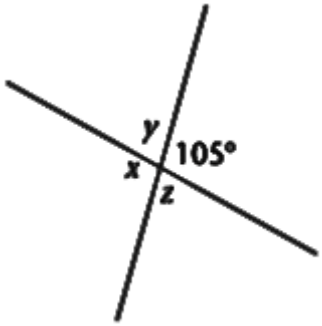
FINAL DRAFT COPY



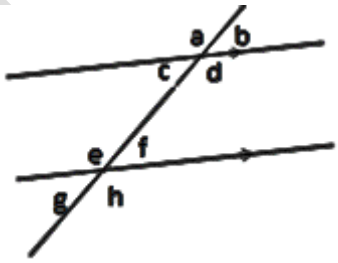
2. Given the lines that intersect below what is the size of angle j? Give reason for your answer.



3. Calculate angle x, y and z by completing the steps below:

	$x = 105^{\circ}$	Vertically opposite angles
	$y+105^{\circ}=180^{\circ}$	angles on a straight line
	$y=180^{\circ}-105^{\circ}$	
	$=75^{\circ}$	
	$y=z$	Vertically opposite angle
	$z=75^{\circ}$	

4. Which set of pairs are:
   (a) Corresponding angles
   (b) Alternating angles
   (c) Co interior angles



5. Study the picture of the house below. Recognize and describe pairs of angles formed by:

FINAL DRAFT COPY

(a) perpendicular lines

(b) parallel lines cut by a transversal

FINAL DRAFT COPY

UNIT 2

GEOMETRY OF 2-D SHAPES

1. Match the properties in Column A with the name of the 2D shape in column B

Column A	Column B
(a) A triangle with three equal sides and angles	i. Isosceles triangle
(b) A triangle with the hypotenuse side	ii. Equilateral triangle
(c) A quadrilateral with all four sides equal	iii. Kite
(d) A quadrilateral with only one pair of opposite sides parallel	iv. Rhombus
(e) A quadrilateral with two pairs of adjacent sides equal and opposite angles between the unequal sides equal.	v. Trapezium
(f) A triangle with two equal sides and two equal angles	vi. Right angled triangle

2. Look at the figures and name each triangle

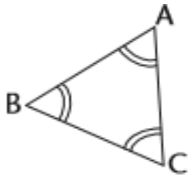


Figure A

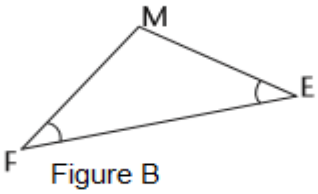


Figure B

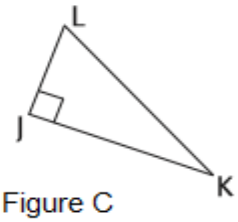


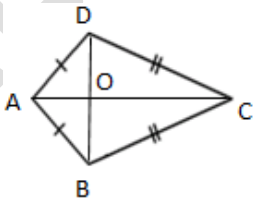
Figure C

3. Identify types of triangles in the following pictures

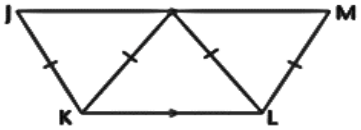
(a)

(b)

4. Divide a kite into triangles and name the triangles

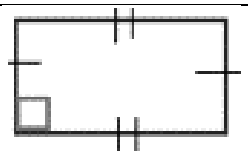
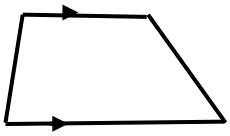
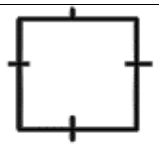
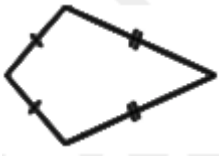
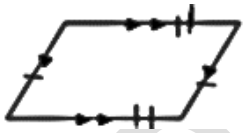


5. Which 2D shapes are found in the diagram below



6. Sort and group the 2D shapes according to their properties.

FINAL DRAFT COPY

A		**B**		**C**	
D		**E**		**F**	

(a) All four sides of equal length

(b) Opposite sides are equal

(c) At least one pair of adjacent sides are equal

(d) Perpendicular sides

(e) All four angles are right angles

(f) Two pairs of opposite sides parallel

(g) Only one pair of opposite sides parallel

7. Name quadrilaterals in number 6 above.
8. Mention 2 properties that are common in all the quadrilaterals.
9. Determine if the following statements are **true** or **false.**

(a) A rhombus is a quadrilateral with all four angles of equal size

(b) A square has four right angles

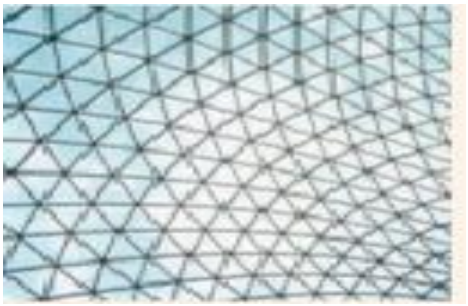
(c) A kite has each pair of opposite angles equal

(d) A rectangle has 2 right angles and 2 acute angles

(e) A parallelogram is a quadrilateral with four angles of different sizes

(f) Total sum of angles in a trapezium is 360˚

10. The quadrilaterals are evident in common structures around the world. Look at the tower and mesh wire below and name the quadrilaterals noticeable.



11. Define parts of the circles

FINAL DRAFT COPY

(a) Centre

(b) Chord

(c) Arc

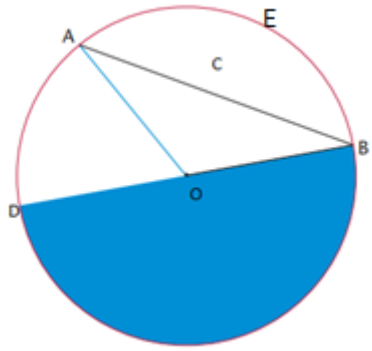
(d) Sector

(e) Segment

(f) Diameter

(g) radius

12. Identify parts of the circle as indicated by letters in the diagram:

	OB AB C Blue area DB O E

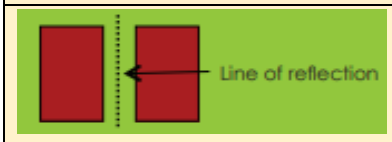
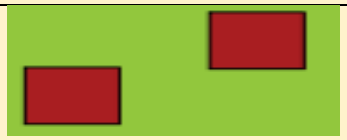
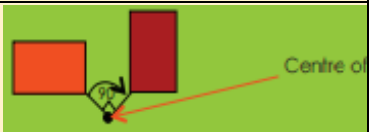
FINAL DRAFT COPY

UNIT 3

TRANSFORMATIONS

Transformation is changing the form of something according to specific rules. A transformation is what brings about the change in position, shape of size of an object. Common transformations are **translations**, **rotations**, **reflections**, and **enlargements**.

Reflection, Transaction and Rotation

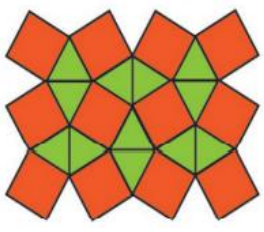
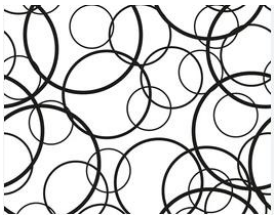
Reflection	Translation	Rotation
A reflection is a transformation that has the same effect as a mirror.	A translation is the movement of an object to a new position without changing its shape, size or orientation.	Rotation: a rotation is a transformation that moves points so that they stay the same distance from a fixed point, the centre of rotation
		

The cited transformation only changes the position of the object. The size does not change, therefore, the image is always congruent to the original figure.

3.1 Use transformations to make composite shapes and tessellations

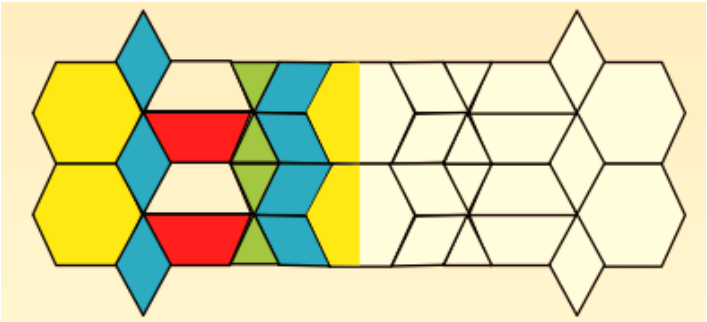
1. What shapes have been used in the patterns below? Are these patterns tessellations?

Tessellation is the patterns formed by shapes, which fit together without any gaps and do not overlap

(a)	(b)	(c)
		

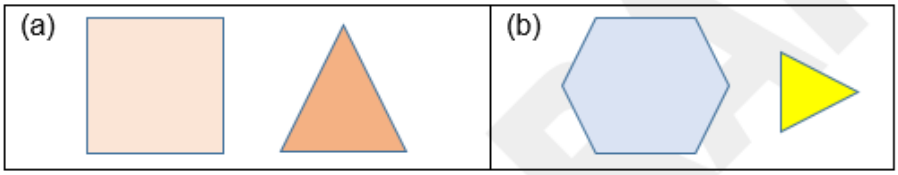
(d)	(e)

2. Complete this reflective tessellation.



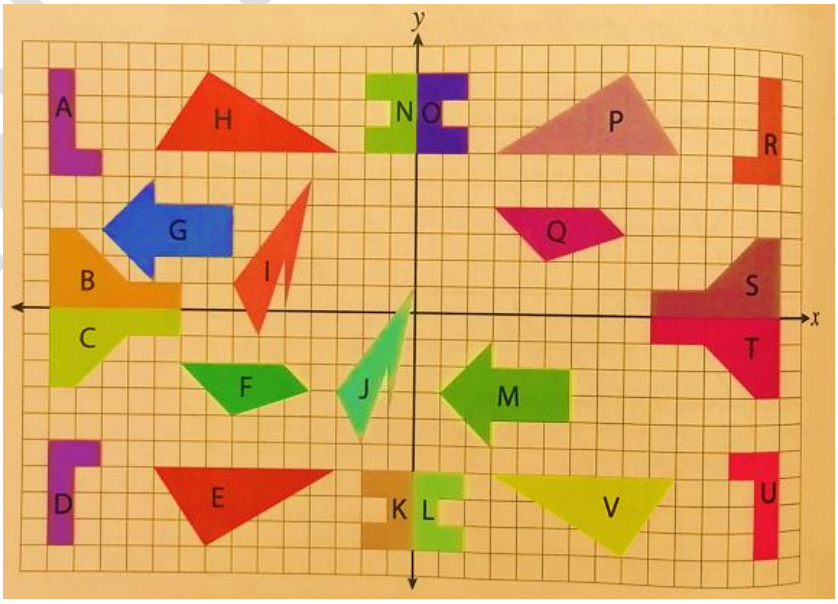
3. Why is this a reflective tessellation?

4. Show a tessellation pattern using the following shapes.



3.2 Recognize, describe and perform translations, reflections and rotations with geometric figures and shapes.

1. Study the following diagram and answer questions.



FINAL DRAFT COPY

(a) List the pairs of shapes that are a reflection of each other.

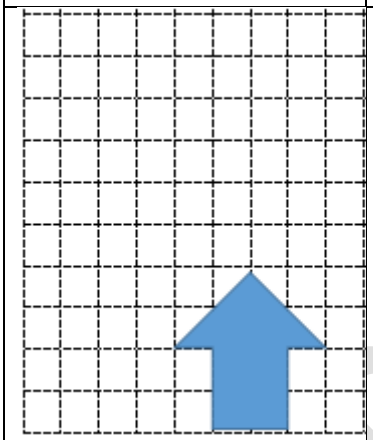
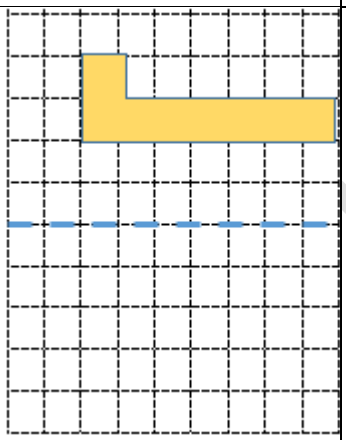
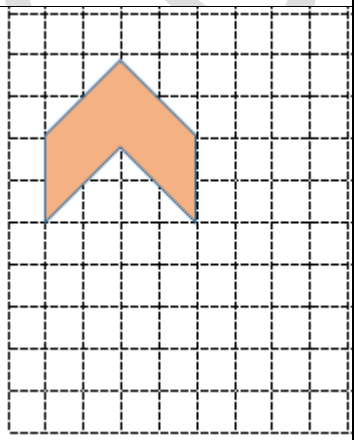
(b) List the pairs of shapes that are a translation of each other.

(c) List the pairs of shapes that are a rotation of each other.

(d) Are there any pairs of shapes that are an enlargement of each other?

(e) Consider shape V and describe the transformations that could be carried out on it.

2. Show the following transformations on a grid.

(a) Translate the shape 2 units to the left and 5 units up.	(b) Reflect the figure below about the line of reflection.	(c) Rotate the shape 90° clockwise.
		

3.3 Enlargement

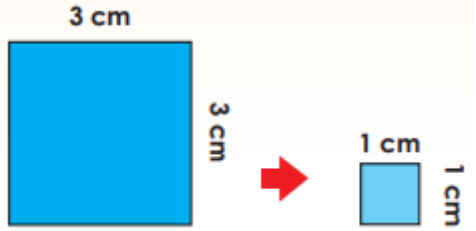
Another important type of transformation is resizing. Resizing can result in a larger shape and that is called enlargement or it can result in a smaller shape and is thus called reduction. If the shape is enlarged proportionally, all their angles are equal and all the sides are increased or decreased in the same ratio. This type of transformation produces similar figures as illustrated.

The scale factor

A **scale factor** is a number by which the size of any geometric figure or shape can be changed with respect to its original size.

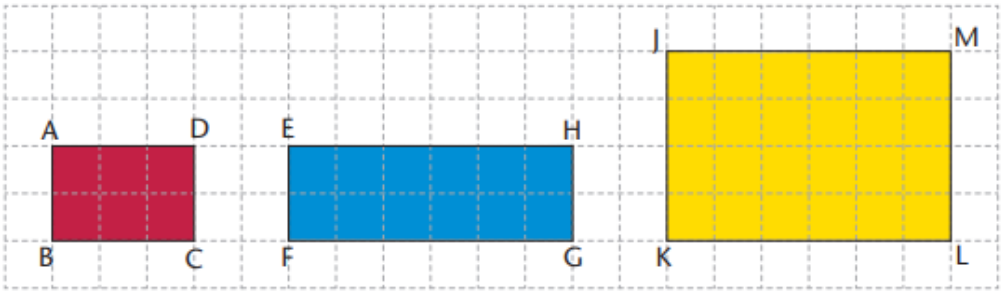
- When the scale factor is 1, the image is the same size as the original shape.

- When the scale factor is less than 1, the image is reduced. For example, if the scale factor is $\frac{1}{2}$ or 0,5, each side of the image is half the length of its corresponding side in the original figure.
- When the scale factor is greater than 1, the image is enlarged. For example, if the scale factor is 2, each side of the image is double the length of its corresponding side in the original figure.



Each dimension of the original shape has been multiplied by $\frac{1}{3}$ in order to get the image, therefore, the scale factor is $\frac{1}{3}$.

3. Look at the following rectangles and answer the questions below.



Rectangle EFGH:

(a) How many times is FG longer than BC?

(b) How many times is EF longer than AB?

(c) Is EFGH an enlargement of ABCD? Give a reason for your answer.

Rectangle JKLM:

(d) How many times is KL longer than BC?

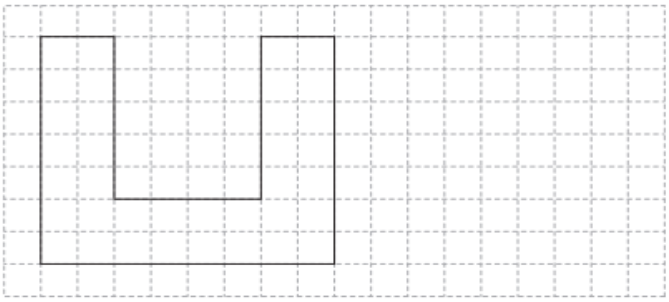
(e) How many times is JK longer than AB?

(f) Is JKLM an enlargement of ABCD? Give a reason for your answer.

FINAL DRAFT COPY

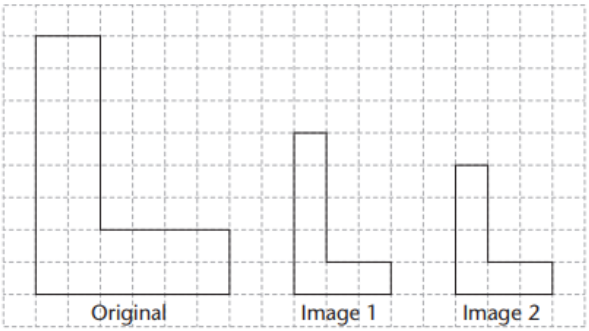
4. On grid paper, enlarge the triangle below with a scale factor of 2.

5. On grid paper, resize the following figure. Use a scale factor of 0,5

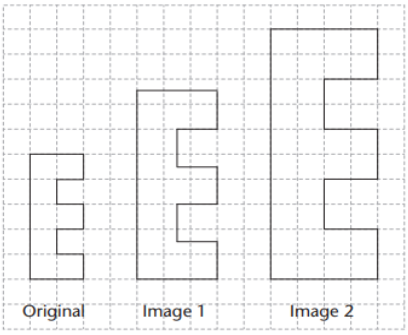


6. Which image is similar to the original shape? State the scale factor by which it has been resized.

(a)



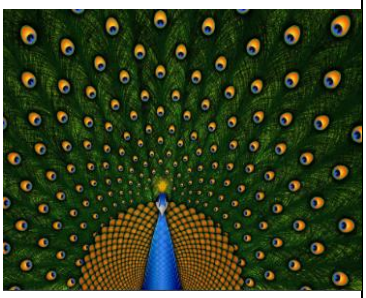
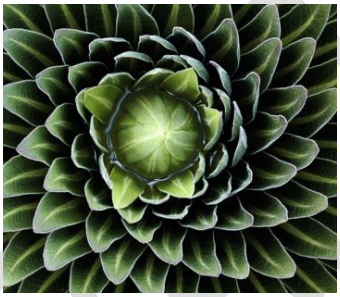
(b)



FINAL DRAFT COPY

3.4 Describing pattern in nature

1. The patterns have been developed using transformation. Describe how each of the patterns were developed.

(a)	(b)	(c)
		
(d)	(e)	(f)
		

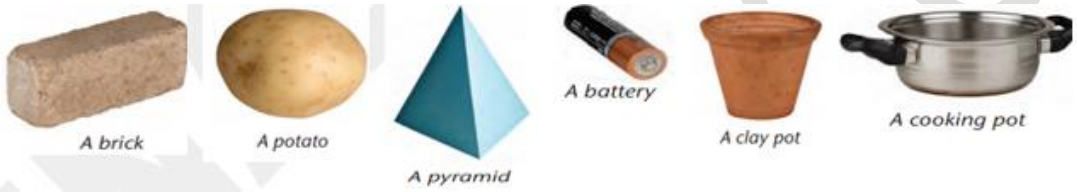
FINAL DRAFT COPY

UNIT 4

GEOMETRY OF 3-D OBJECTS

Most objects we see around us, like fruit, animals, trees, rocks, people and motor cars, have curved or round surfaces. Some objects, like a saucepan or other cooking vessel, have both round and flat surfaces. Other objects only have flat surfaces.

1. Study the following objects.



(a) Which objects have curved surfaces only?

(b) Which objects have flat surfaces only?

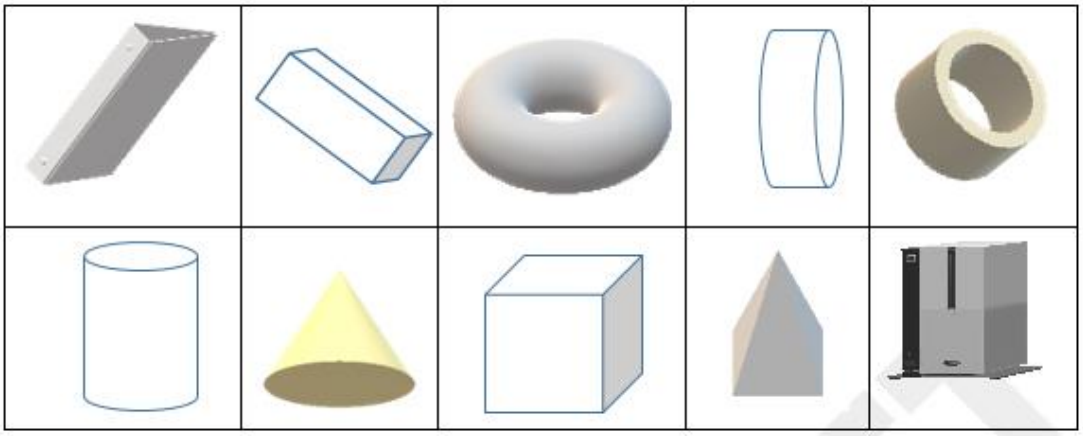
(c) Which objects have both curved and flat surfaces?

(d) Which of the objects has the shape of a sphere?

(e) Which of the objects have the shape of a cylinder?

(f) Which of the objects have the shape of a rectangular prism?

2. Study the following diagram and answer questions.



(a) Shade objects with curved surfaces only yellow.

(b) Shade objects with flat surfaces only, green.

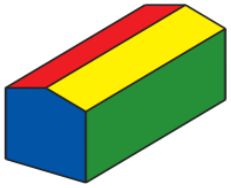
(c) Shade the objects with both curved and flat surfaces red.

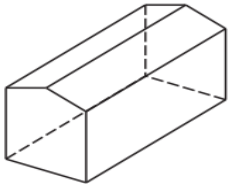
(d) Which of the objects are polyhedra?

4.1 Describe, sort and compare polyhedra

A 3D object with flat surfaces (faces) only is called a polyhedron. A straight edge is formed where two flat surfaces meet. The point where two or more edges meet is called a vertex.

1. In the drawing of the same box on the right, dotted lines are used to indicate edges and surfaces that are hidden in the coloured drawing.





(a) How many faces (different flat surfaces) does this object have altogether?

(b) How many faces cannot be seen in the coloured drawing on the left?

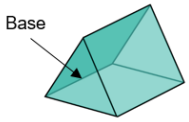
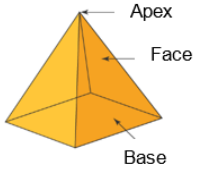
(c) How many of the faces are rectangles?

(d) How many of the faces are pentagons?

(e) How many edges does the coloured polyhedron have?

(f) How many vertices does it have?

There are two special groups of **polyhedra i.e. prisms** and **pyramids**.

Prisms	**Pyramids**
Prism are polyhedra with two faces that are congruent and parallel polygons. These two faces are called bases and are connected by lateral faces that are parallelograms. Prisms are named using their base. The two opposite congruent faces are the base. **Example:** Triangular prism 	Pyramid have one base. The lateral faces of a pyramid are always triangles. These triangles meet at the same vertex. This vertex is called the **apex** of the pyramid. Pyramids are named using their base. The base is not always the face it is sitting on. **Example:** Square-based pyramid 

FINAL DRAFT COPY

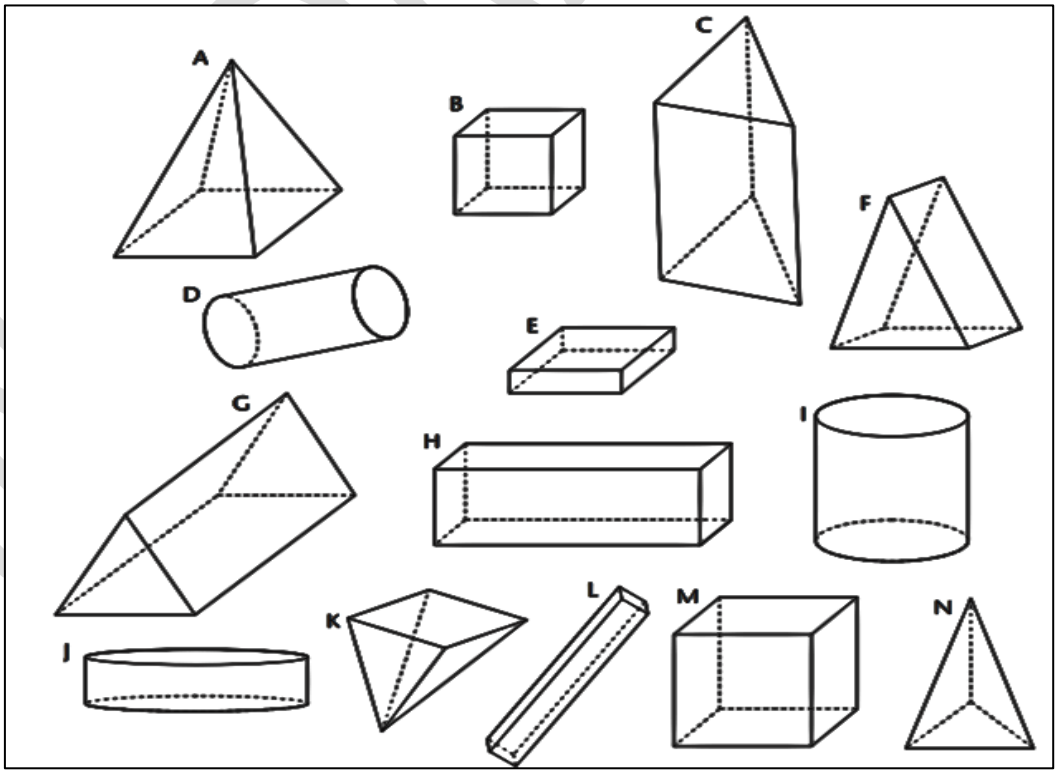
1. Copy and complete the following table of **prisms**.

No. of sides of a base	Sketch	Name of the object	No. of faces	No. of vertices	No. of edges
3					
4					
5					
6					

2. Copy and complete the following table of **pyramids**

No. of sides of a base	Sketch	Name of the object	No. of faces	No. of vertices	No. of edges
3					
4					
5					
6					

3. Study the following 3D objects.



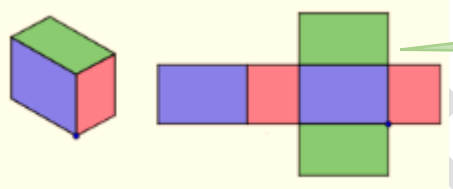
FINAL DRAFT COPY

(a) Sort the 3D objects below into the three groups, namely prisms, pyramids and cylinders. Write down the correct letter under each group.

(b) Sort the prisms into three groups (cubes, rectangular prisms and triangular prisms). Write the letter under each group.

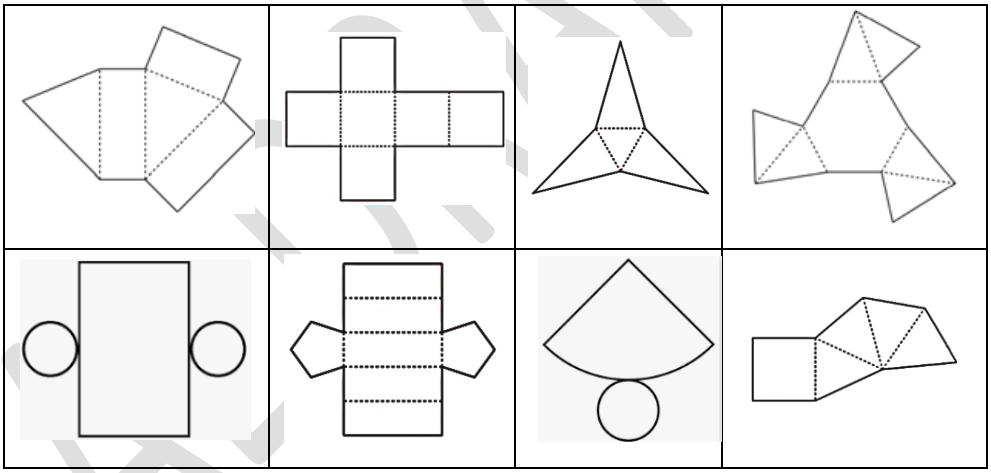
4.2 Building 3-D models using nets

In mathematics, a **net** is a flat pattern that can be folded to form a 3D object. Different 3D objects have different nets. Sometimes the same 3D object can have different nets.

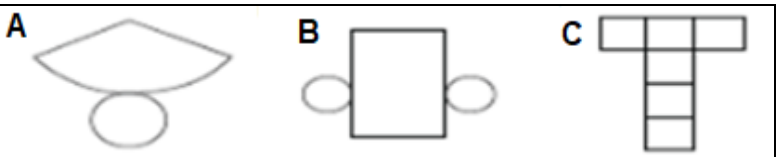


Net of a rectangular prism

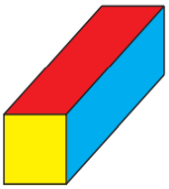
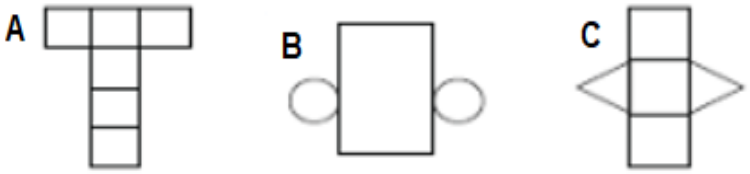
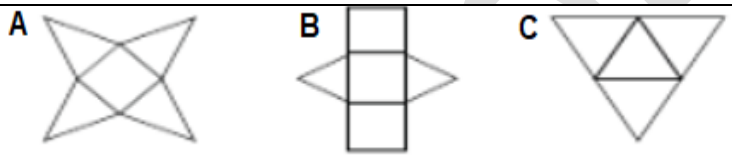
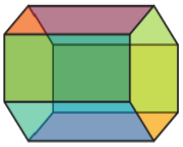
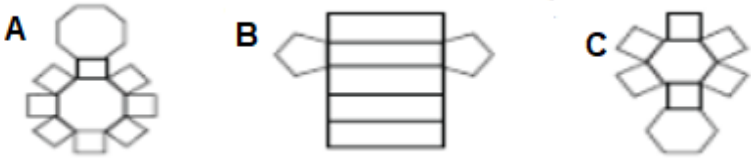
1. Make geometric objects using the nets provided. You will need some sticky tape and a pair of scissors. Examples of nets have been provided but more nets can be added.



2. Choose the matching net for each geometric solid. Write the number and corresponding letter of alphabet.

I.		A	

FINAL DRAFT COPY

II.			
III.			
IV.			
V.			

FINAL DRAFT COPY

UNIT 5

AREA AND PERIMETER

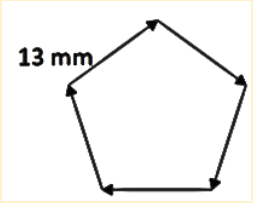
5.1 Calculate the perimeter of regular and irregular polygons

The perimeter of a polygon is the total distance around the polygon.

Regular polygons

Regular polygons have all sides and angles equal. The perimeter of regular polygons is found by adding all the sides of a regular polygon or multiplying one side of the regular polygon by the number of its sides.

The pentagon below is an example of a regular polygon. Its perimeter is found by multiplying the 13 mm by 5.



Perimeter(P) $= 5 \times 13$ mm

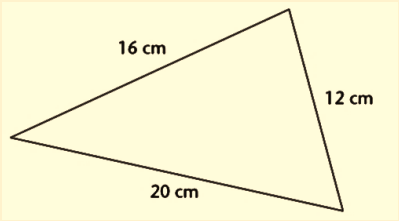
$= 65$ mm

Irregular polygons

A polygon is said to be irregular when its sides are not equal to each other, and the measure of angles may also be unequal. The perimeter of irregular polygons is found by adding all the sides of the irregular polygon.

Example

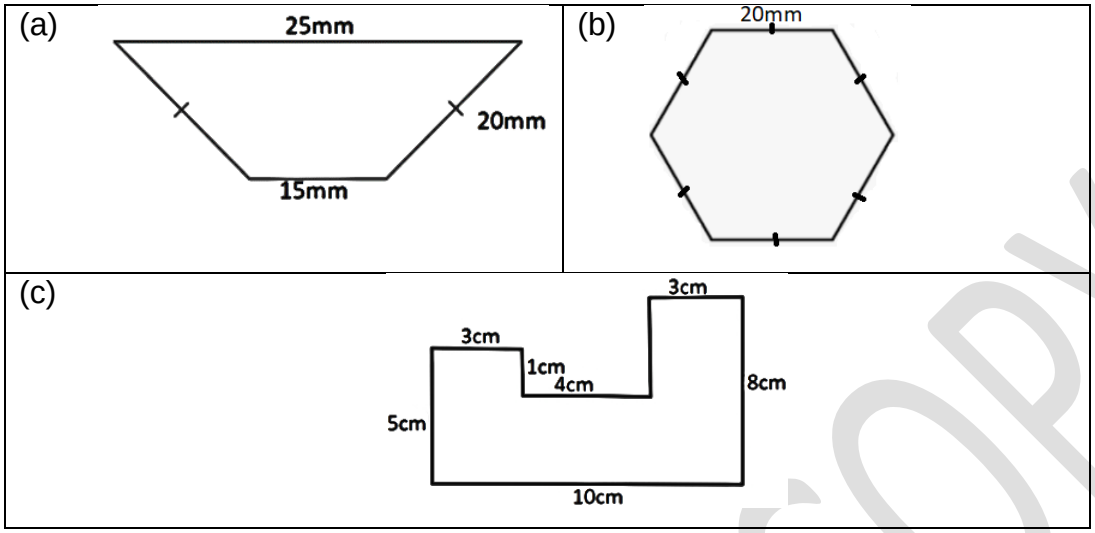
Calculate the perimeter of the following shape



P $= 16$ cm $+ 12$ cm $+ 20$ cm

$= 48$ cm

1. Calculate the Perimeter of the polygons below using appropriate formulae. Leave your answer in centimetres.

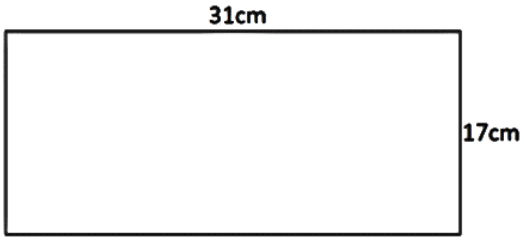


5.2 Use appropriate formulae to calculate the perimeter of squares and rectangles

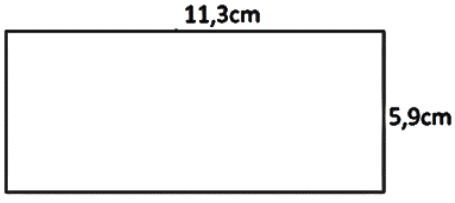
- To calculate the perimeter of a square, we add all the sides of a square. The formula to calculate the perimeter of a square is **P = 4s** (where s represents side).
- The formula to calculate the perimeter of a rectangle is **P = 2*l* + 2 *b* or P = 2 (*l* + *b*)**.

1. Calculate the perimeter of the following quadrilaterals with the given lengths and breadths. Leave your answer in centimetres.

   (a)

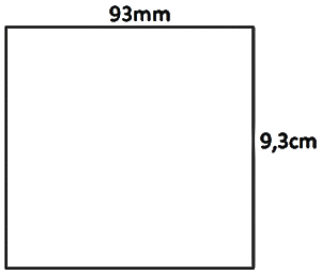


   (b)



   (c)

FINAL DRAFT COPY

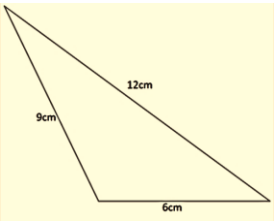


(d) If the perimeter of the square is 480 mm. What is the length of each side?

5.3 Use appropriate formulae to calculate the perimeter of triangles and circles

To calculate the perimeter of a triangle, we add all the sides of a triangle. The formula to calculate perimeter of a triangle is **$P = s_1 + s_2 + s_3$**

Example



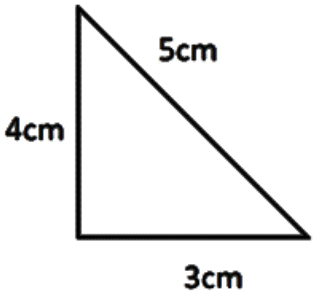
$P = s_1 + s_2 + s_3$

$= 12\text{ cm} + 9\text{ cm} + 6\text{ cm}$

$= 27\text{ cm}$

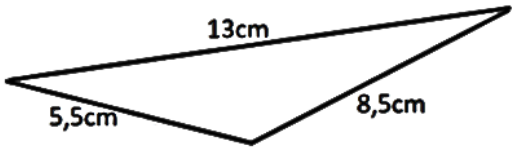
1. Calculate the perimeter of the following triangles given the lengths of the sides.

(a)

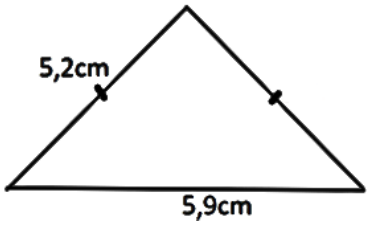


(b)

FINAL COPY



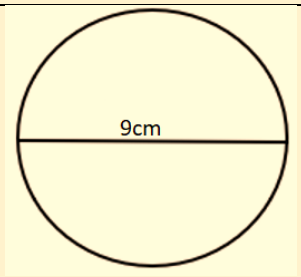
(c)



To calculate the perimeter of a circle, we take the whole distance around the circle. The circumference of a circle is the same as its perimeter. The formula to calculate the perimeter of a circle is **P = 2πr**

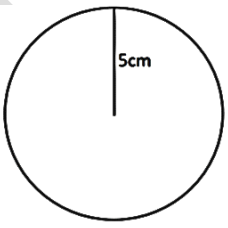
Example

Calculate the perimeter of the given circle.

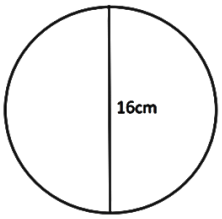
	P $= 2\pi r$ $= 2\pi(4{,}5\text{ cm})$ $= 2 \times 3.14 \times 4{,}5\text{ cm}$ $= 28.26\text{ cm}$

2. Calculate the perimeter of a circles below:

(a)



(b)



(c) What is the perimeter of the circle with a diameter of 19 cm?

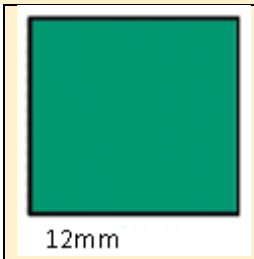
FINAL DRAFT COPY

(d) What is the circumference of a pizza if one straight side of each slice equals 3,1 cm?

5.4 Use appropriate formulae to calculate the area of squares and rectangles

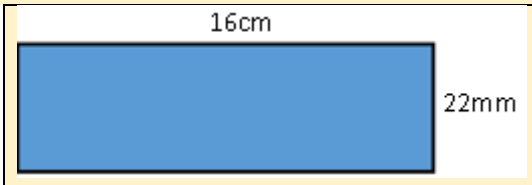
Area of the square

Area of the square is calculated by the formula: $A = l \times l = l^2$

	$A = l \times l$ $= 12\text{ mm} \times 12\text{ mm}$ $= 144\ mm^2$

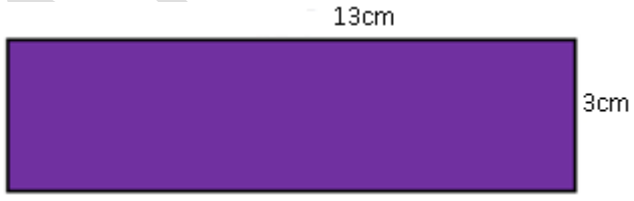
Area of a rectangle

Formula to calculate Area of a rectangle is $A = l \times b$

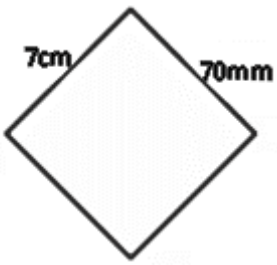
	$A = l \times b$ $A = 16\text{ cm} \times 22\text{ mm}$ $= 16\text{ cm} = 2{,}2\text{ cm}$ $= 35{,}2\ cm^2$

1. Calculate the area of the following rectangles and squares

(a)



(b)

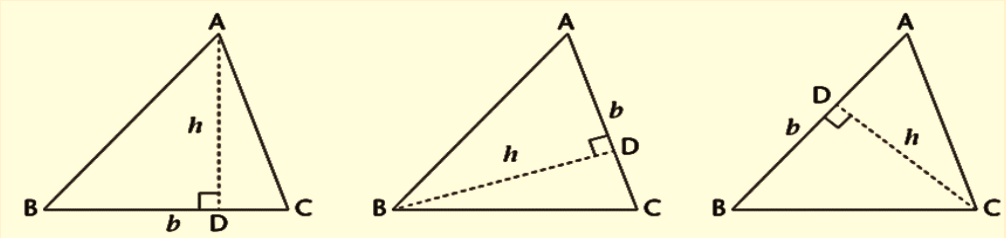


(c) Given the area of a square 121 m^2 calculate its length.

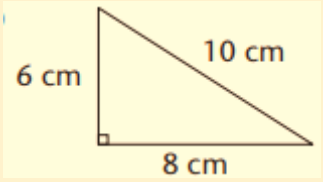
(d) If the area of a rectangle is 200 cm^2, and its length is 50 cm, what is its width.

5.5 Use appropriate formulae to calculate the area of triangles and circles

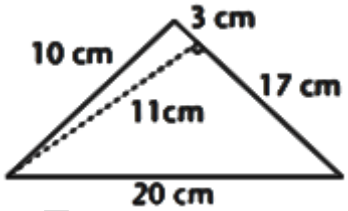
The formula to calculate area of a triangle is $\mathbf{A} = \frac{1}{2} \times \boldsymbol{b} \times \boldsymbol{h}$ (where *b* represents the base and *h* the height). The height is a line segment drawn from the vertex perpendicular to the base.



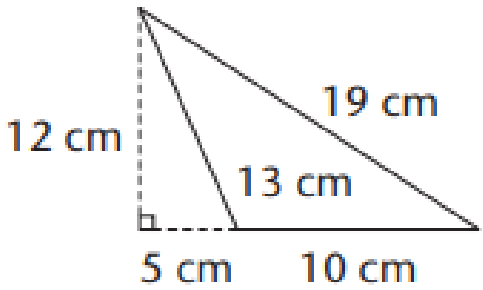
AD = height	BD = height	DC = height
BC = base	AC = base	AB = base

	$\mathbf{A} = \frac{1}{2} \times \boldsymbol{b} \times \boldsymbol{h}$ $\mathbf{A} = \frac{1}{2} \times \mathbf{8\ cm} \times \mathbf{6\ cm}$ $= \frac{1}{2} \times \mathbf{48}$ $= \mathbf{24}\ \boldsymbol{cm^2}$

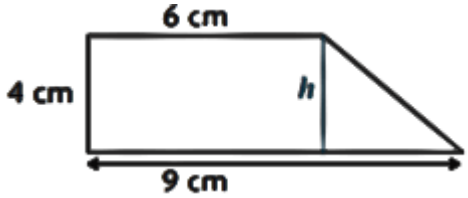
a) Calculate area of the given triangle.



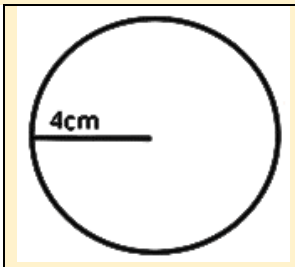
b) Calculate the area of the triangle below:



c) Work out the length of the sides you need and then calculate the area of the shape.



The formula to calculate the area of circles is $\boldsymbol{A = \pi r^2}$ (where r is the radius of the circle and $\boldsymbol{\pi = 3.14}$). The radius of the circle below is 3cm. The area of the circle is:

	A= πr^2 = 3,14(4cm)^2 = 3,14 ×16 cm^2 = 50,24 cm^2

The formula to calculate area of a circle with a given diameter is $\mathbf{A = \pi(\frac{d}{2})^2}$

NB Diameter of a circle is double its radius.

1. Calculate the area of circles with the following:
   (a) Radius of 83 cm
   (b) Diameter of 257 mm
2. Calculate the circumference of a circle with
   (a) Radius of 2 cm
   (b) Diameter of 25 mm.

5.6 Problem solving involving perimeter and area of polygons

1. A man wants to put a fence around a piece of land to keep his sheep away from his cows. One side fence 6m, the second 7m, the third 8m and the fourth 10m. What is the total perimeter of the fence?
2. You need to tile the floor of your room with an area of 4,2m x 3,5m. The tiles you want to buy are 45cm x 45cm. How many tiles do you need?
3. Mandla draws a circle with a diameter of 16cm. He colours one half of the circle.
   What is the area of the shaded part?

FINAL DRAFT COPY

UNIT 6

SURFACE AREA, VOLUME AND CAPACITY

6.1 Using appropriate formulae to calculate the surface area, volume and capacity of cubes

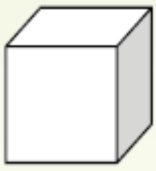
(DBE Workbook 1 - Grade 7, Page 131

Example:
The formula for the volume of a cube is ℓ^3

2 cm × 2 cm × 2 cm
= 8 cm³
= 8 ml
= 0,008 ℓ

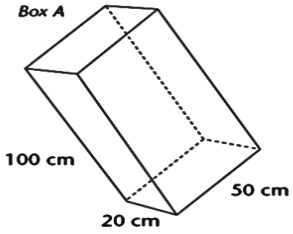
1. Calculate the total surface area of the given cube and give the answer in cm^2 and m^2

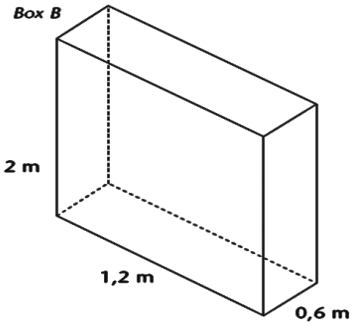
2. Use the formulae to calculate the volume and capacity of water that will fill the following cube.



6.2 Using appropriate formulae to calculate the surface area, volume and capacity of rectangular prisms.

1. The two boxes on the following page are rectangular prisms. The boxes must be painted.
   (a) Calculate the total surface area of Box A and of Box B.



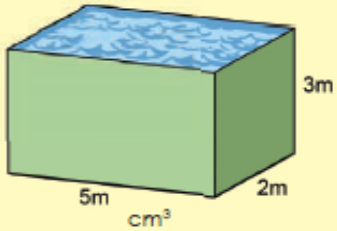


FINAL DRAFT COPY

(b) What will it cost to paint both boxes if the paint costs R1,34 per square metre?

(DBE Workbook Grade 7, Page 129

Example:



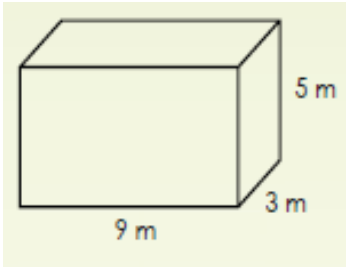
This container will hold 30 000 000 ml or 30 000 ℓ of water.

$\frac{m^3}{l \times b \times h}$
$= 5\ m \times 2\ m \times 3\ m$
$= 30\ m^3$

$\frac{cm^3}{l \times b \times h}$
$= 500\ cm \times 200\ cm \times 300\ cm$
$= 30\ 000\ 000\ cm^3$

$\frac{mm^3}{l \times b \times h}$
$= 5\ 000\ mm \times 2\ 000\ mm \times 3\ 000\ mm$
$= 30\ 000\ 000\ 000\ mm^3$

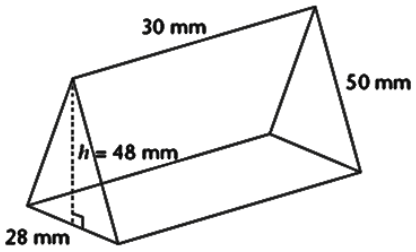
2. Calculate the volume of the following and give your answer in m^3, cm^3 and mm^3.Also say what the capacity of each container is when filed with water.



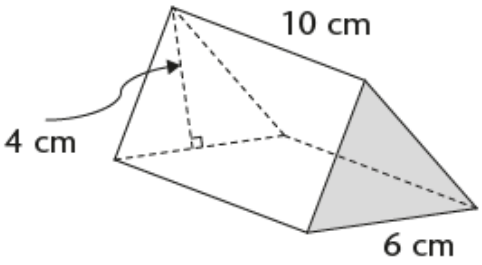
6.3 Using appropriate formulae to calculate the surface area, volume and capacity of triangular prisms.

1. Work out the total surface areas of the following objects. Give all the answers in cm^2.

(DBE Textbook- Grade 8 LB, Pg 78)



2. Calculate the volume and capacity of the following objects using the formulae.



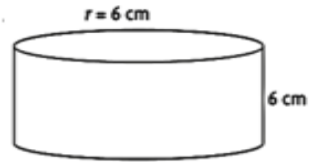
FINAL DRAFT COPY

6.4 Using appropriate formulae to calculate the surface area, volume and capacity of cylinders.

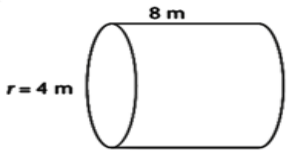
1. Calculate the surface areas of the following objects. Use π = 3,14 and round off all your answers to two decimal places.

DBE Textbook LB Gr 9 page 80)

(a)



(b)

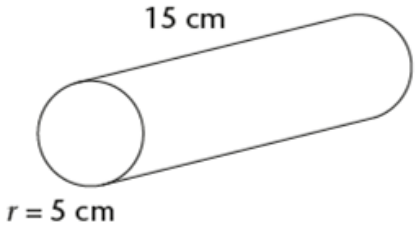


2. Calculate the volume and capacity of the following cylinders. Use π = 3,14 and round off all answers to two decimal places.

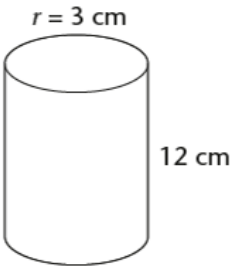
Volume of a cylinder = Area of base × h = $\pi r^2 \times h$

(a)

DBE Textbook LB Gr 9 page 83)



(b)



6.5 Solving problems in context involving.

1. A rectangular – shaped petrol tank is used to store a reserved petrol. The tank has a side length of 8m, width of 3m and a height of 4m.

(a) What is the capacity of the tank? Write your answer in litres

(b) If the tank is quarter full of petrol, what does it hold in kilolitres?

FINAL DRAFT COPY

UNIT 7

DATA HANDLING

Data handling refers to the process of collecting the set of information, organize and represent in different forms. Data can be about many different things, for example, people's opinions on politics or the success rates of treating people with a certain kind of medicine. We use data to help us make decisions and solve problems about the world around us.

7.1 Collect and organize data

Collecting Data

Data collection is a systematic process of gathering and measuring information. A questionnaire with suitable multiple choice questions can be a tool to collect the desired information.

The method of collecting data will help us to identify the most appropriate tool to be used when collecting the data. There are various sources of data collection.

Multiple-choice Questionnaire

Year 4 learners are involved in a recycling project. You want to determine which type of material they are mostly bringing to school. You want to investigate this by means of a survey. A multiple-choice questionnaire can be used to record your data.

1. Design a simple questionnaire for your survey, using multiple-choice questions. Your questionnaire must also include:
   (a) Gender
   (b) Type of material
2. Use the multiple-choice question that you developed in Question (l) to collect the data from 60 learners (30 boys and 30 girls). Keep the results for the next section.

Organising data

Organizing data is a systematic way of bringing data together in a way that makes it easier to read. Tally marks (l) can be used to record data in different

categories as shown in the callout. Tallies are grouped in to five to make it easier to count them quickly.

Tally marks
A count of three = 𝍸
A count of five = 𝍸𝍸
A count of seven = 𝍸𝍸 𝍸

3. Use the information you have collected on recycling material per gender to construct a tally table.
4. Portia's collected data is organized in the table below:

Recycling material / Gender	Plastic bags	Paper	Tins	Bottles	Boxes
Girls	2	10	12	5	1
Boys	5	3	6	12	4

Complete the table using Portia's data

Recycling material	Tally (total number of learners)	Frequency (total number of learners)
Plastic bags		
Paper		
Tins		
Bottles		
Boxes		

5. The table below represents the quantity of recycling material collected Year 4 learners. Construct a frequency table for this set of data values, using grouped data, grouped in five groups with intervals of 10.

58	68	60	71	53	62	64	72
63	46	61	52	67	54	63	78
78	62	68	55	69	81	76	62
52	64	65	74	59	66	63	76
55	74	74	59	51	59	67	71

FINAL DRAFT COPY

7.2 Summarizing Data

Measures of central tendency and measures of spread and dispersion

A measure of central tendency is a single value that attempts to show a central position of a set of data. There are three types of measures of central tendency i.e. mode, median and mean.

Mode

The mode is the data value that appears most often in a set of data. No calculation is needed to find the mode. You just find the value that appears most frequently. If no number appears more than the other numbers, then there is no mode. It is important to arrange data in ascending or descending order in order to easily identify the mode.

Median

The median is the middle number in the data set. To determine the median, you have to write all the numbers in the data set from the smallest to the highest and the number in the middle will be your median. If there is more than one number in the middle (i.e. if the data contains an even number of data values) add the two numbers in the middle and divide the answer by 2.

Mean

The mean or the average is the most common measure of central tendency that is used, but it can be easily influenced by high or low numbers in the data set. It is calculated by adding all the values together and dividing by the number of values in the data set.

Range

The range of a data set is the difference between the highest value and the lowest value. For example, the range of the data 18; 12; 10, 10, 11 is: $18 - 10 = 8$.
The bigger the range, the more the data is spread out. The smaller the range, the more the data is clustered around similar values

1. Use the data you collected (about recycling materials) for both boys and girls to determine the mode, median, mean and range.

   Set of scores:

Mode		
Median		
Mean		
Range		

2. Use the combined data from Portia's findings to determine the mode, median, mean and the range

Portia's collected data:

Recycling material / Gender	Plastic bags	Paper	Tins	Bottles	Boxes
Girls	2	10	12	5	1
Boys	5	3	6	12	4

Mode	
Median	
Mean	
Range	

3. Use the following data set to calculate the range, mode, median and the mean.

   The table represents times in seconds taken by the different learners in Year 3 in the same school, to wash their hands:

20	30	36	14	20	14	29	39	15	37	35	24	29
29	18	16	38	13	24	27	22	38	29	11	38	

4. The table represents times in seconds taken by the different learners in Year 4 in the same school to wash their hands:

20	12	39	22	16	37	36	15	14	13	16	10	14
13	16	37	36	15	14	13	16	10	14	26	11	

FINAL DRAFT COPY

(a) Calculate the range for each set of data

Year 3: _____ Year 4: _____

(b) What can you say about the two groups by looking at the range?

(c) Calculate the mean (average) for each group

Class	Total	Number	Mean
Year 3			
Year 4			

(d) Compare the two set of data using the mean

_____

(e) Find the median for each class

Class	Scores from highest to lowest	Middle position	Median
Year 3			
Year 4			

(f) Compare the two sets of data using the median

(g) Find the mode for each class

Class	Highest frequency	Mode
Year 3		
Year 4		

(h) Compare the two sets of data using the mode

7.3 Represent data

Data can be represented using different types of graphs. They help with displaying and reporting data and make it easier to report patterns and relationships. We will be working with three graphs i.e. bar graphs, double bar graphs, histograms, pie charts and broken line graph.

Bar graph

A bar graph usually shows categories (or classes) of data along the horizontal axis, and the frequency of each category along the vertical axis.

FINAL DRAFT COPY

A bar graph is a visual display used to compare the frequency of occurrence of different characteristics of data.

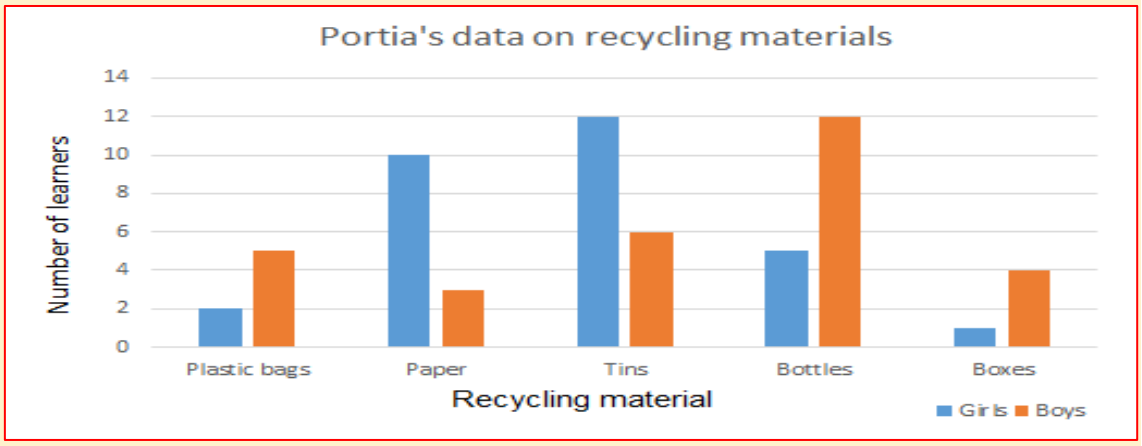
Double bar graph

A double bar graph shows two sets of data for each category (or class).

Example

Use Portia's frequency table to represent data in a double bar graph.

Recycling material / Gender	Plastic bags	Paper	Tins	Bottles	Boxes
Girls	2	10	12	5	1
Boys	5	3	6	12	4



1. Use your frequency table on recycling materials per gender to represent data in a bar graph (only girls)

Histogram

2. The learners from Year 1 to 4, at school B were asked how much time (in minutes) did they spend watching videos in you tube. The following table shows the results from a sample of 150 learners:

Time in minutes	Frequency
0 – 10	19
11 – 20	52
21 – 30	38
31– 40	18
41 – 50	10

51 – 60	6
61 – 70	7

(a) In which interval are most learners on you tube?

(b) Draw the histogram to illustrate the data

(c) Describe in your own words what the histogram shows.

(d) What would you expect this data to look like for a school in a farming area?

Pie chart

A pie chart is a circular graph in which the circle is divided into sectors. Each sector visually represents an item in a data set to match the amount of the item as a percentage or fraction of the total data set. (the whole circle) Pie charts are useful to compare different parts of a whole amount. They are often used in media to show clear and important differences, but they cannot show shape and spread of data

3. Use your frequency table on recycling material per gender to represent your data in a pie chart (only boys)
4. Use Portia's frequency table to represent the data in a pie chart (total number of learners in each age group)

Recycling material / Gender	Plastic bags	Paper	Tins	Bottles	Boxes
Girls	2	10	12	5	1
Boys	5	3	6	12	4

Broken line graph

A broken-line graph is a line that joins consecutive data points plotted on a set of axes. It is useful to show how something has changed or stayed the same over time, and where the data values vary considerably as in the example where the rainfall goes up and down. Meteorologist use line graphs to show monthly rainfall. Businesses often use it to show information about profits.

With some line graphs it might be possible to continue the line to show what might happen in the future.

FINAL DRAFT COPY

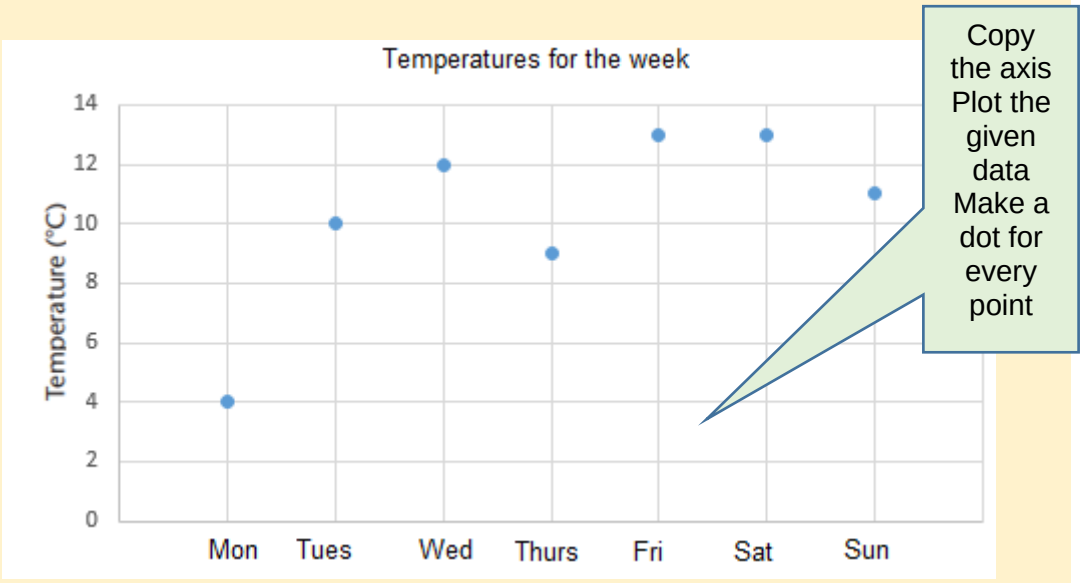
Steps to draw the broken line:

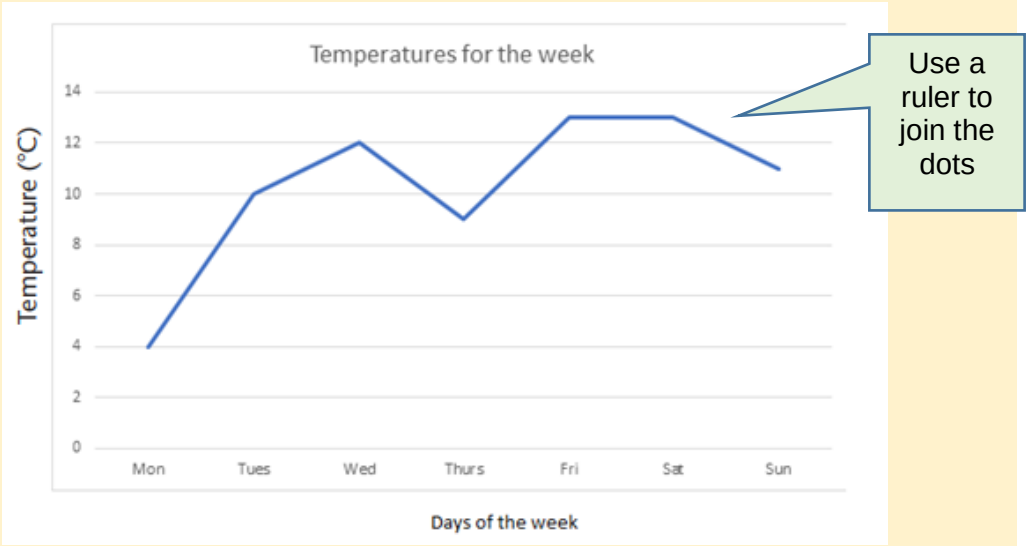
- Copy the axes.
- Plot the given data on the set of axes.
- Make a dot for every point that you plot
- Use a ruler to join the dots in order. You now have drawn a broken-line graph

Example:

The following table shows the average temperature in Mahikeng recorded every day for one week:

Day	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
Temperature (°C)	4	10	12	9	13	13	11

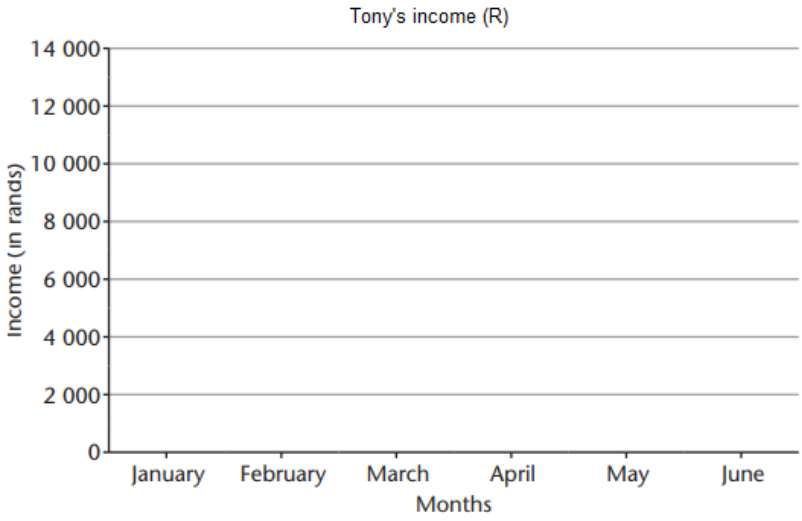




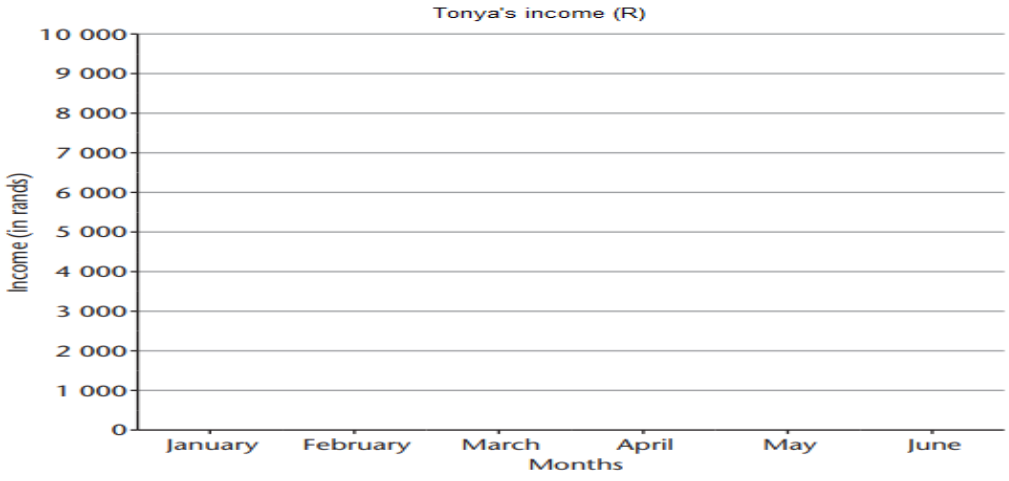
5. Search for a broken line graph in a newspaper or the internet. Redraw it and then describe it.
6. The table below shows the income of Tony's small business and Tonya's small business over six months:

Month	January	February	March	April	May	June
Tony's income (R)	12 000	12 000	9 000	6 000	7 000	9 000
Tonya's income (R)	6 000	7 000	8 000	8 000	9 000	9 000

(a) Copy the axes below and draw a broken-line graph showing Tony's income



(b) Copy the axes below and draw a broken-line graph showing Tonya's income



FINAL COPY

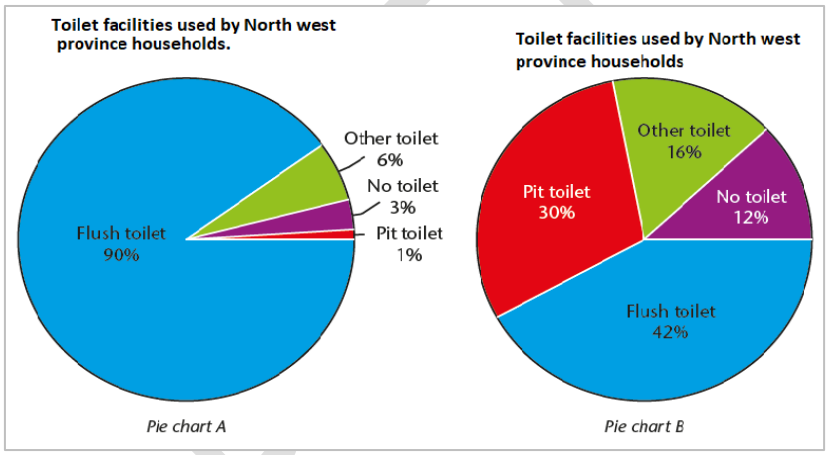
7.4 Interpret and reporting

Data **analysis** and **interpretation** is a process of assigning meaning to the collected information and determining the conclusion, significance and implications of the findings. There are four steps to data interpretation i.e. collect data required; develop findings; develop conclusions and develop recommendations.

There are four steps to data interpretation.

- Collect the information you need.
- Develop findings
- Develop conclusions
- Develop recommendations.

1. The pie chart below represent data collected about toilet facilities in different years in households in North West province.



(a) According to pie chart A, what type of toilet facility do most people have and what percentage of households is this?

(b) How will your answer for (a) be different if you use pie chart B to answer the question?

(c) Write a short report in one paragraph about the data in the pie charts.

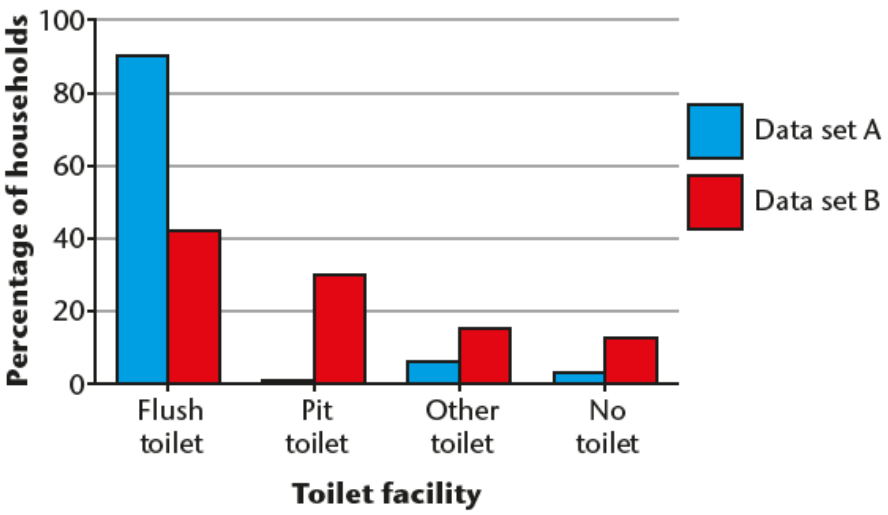
(d) What can you conclude from the data in pie chart A?

(e) What can you conclude from the data in pie chart B?

(f) How could the year of data collection account for the differences in the data?

2. The graph below shows the same data as the two pie charts.

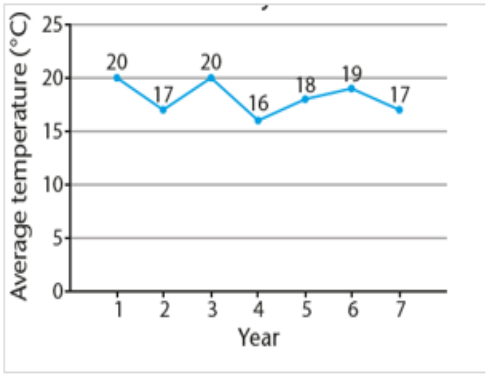
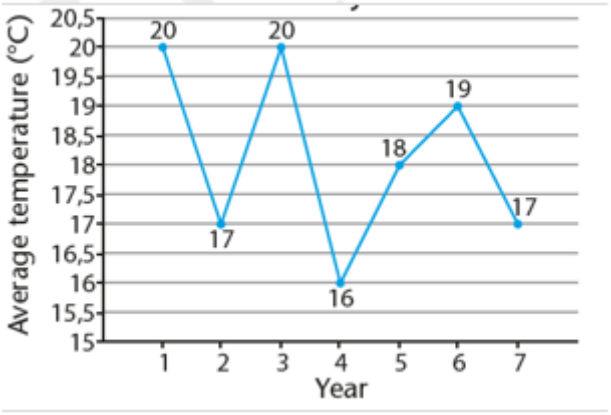
Toilet facilities used by North west province households



(g) Which graph allow us to compare the two sets of data more easily? Motivate your answer

(h) Which type of graph do you prefer to represent this type of data? Why?

3. The graphs represent the average temperatures recorded in the same place, at the same time over a period of seven years.

Graph A: Average temperature over 7 years	**Graph B: Average temperature over 7 years**
	

(a) Do both graphs show exactly the same data?

(b) Why do the graphs look so different?

FINAL DRAFT COPY

(c) Which of the graphs would people use to emphasise that there are big differences in the temperatures over the years? Explain your answer

(d) Suggest a way to change the vertical scale of Graph A to emphasise even more that there are no big differences between the temperatures over the years.

(e) Write a short report on Graph A. Also include a prediction of the temperatures in years 8 and 9.

7.5 Reporting

To report data one needs to consider the following:

Plan: What data do you need? Who will you get it from? How will you collect it? How will you record it? How will you make sure the data is reliable? Give reasons for the choices you made.

Analysis: This is where you do the calculations and draw charts.

- Compare groups with the mean and median.
- The range is a measure of how spread out the group is.
- Graphs are good for representing visual data.

Conclusions: Do your results agree with the hypothesis? How confident are you? What went wrong? How did you deal with it? What would you do differently if you did the research again?

1. Read the following paragraph and answer the questions that follows:

   A recent study revealed that 35% of Year 4 learners smoke cigarettes, 49% drink alcohol and 8% abuse drugs. This is an indication of the general poor health and social problems of the teenage population in our country.

   (a) Do you agree that the figures are high enough to conclude that the habits of these teenagers are unhealthy?

   (b) Can we tell the following from the data above?

   - What was the sample of this study?
   - Where was the data collected?
   - When was the data collected?

FINAL COPY

(c) If the sample consisted of ten teenagers all located in an area known for drug and alcohol abuse, would the data be a good reflection of all teenagers in the country?

(d) Describe what you think would be a better sample.

(e) Why is it important to know the date when this data set was collected?

2. Hypothesis: Girls prefer Accounting and Maths above Physical Sciences and Social Sciences.

Name	Favourite subjects	Name	Favourite subjects
Thabo	Accounting	Victoria	Mathematics
George	Life sciences	Lungile	Mathematics
Rorisang	Mathematics	David	Accounting
Andile	Accounting	Kate	Geography
Tshepo	Mathematics	Nelly	Mathematics
Wonderboy	Geography	Elizabeth	Physical Sciences
Modise	Physical Sciences	Peter	Life Sciences
Rebaona	Accounting	Themba	Accounting
Lerato	Life Sciences	Kabelo	Physical Sciences
Faith	Mathematics	Rejoice	Geography

Use the data set above, and write a report on your findings. Include your frequency table, graphs and conclusions. Also compare the favourite subjects of girls to those of boys.

FINAL DRAFT COPY

TERM 4

FINAL DRAFT COPY

Contents

TERM 4
Table of Content

Unit 1
Whole Numbers……103
1.3 Addition and Subtraction……..……103
1.4 Multiplication……64
1.5 Ration and rate……104

Unit 2
Fraction
2.1 Common
Fraction……105
2.2 Decimal
Fraction……..107
Unit 3
Algebra Language……110
Unit 4
Numeric and Geometric Patterns……113
Unit 5
5.1 Length……118
5.2 Mass……118
Unit 6
Construction of Geometric Figures……120
Unit 7
Geometry of Straight Lines……..123
Unit 8
Geometry of 2-D Shapes……..126

FINAL DRAFT COPY

FINAL DRAFT COPY

UNIT 1

WHOLE NUMBERS

1.1 Addition and Subtraction

1. Calculate the following using any method and verify your answers using the calculator:

   (a) 584 364 + 417 120 =

   (b) 734 249 + 46 342 =

   (c) 563 482 – 278 356 =

   (d) 735 593 – 149 256 =

1.2 Multiplication and Division

1. Write the first 5 multiples of:

(a) 7

(b) 9

(c) 15

2. Write down the factors of:

(a) 28

(b) 26

(c) 60

Multiplication: Use a range of techniques to perform written calculations with whole numbers

3. Multiply the following numbers and check your answers using a calculator.

(a) 4 625 × 213 =

(b) 2 417 × 124=

(c) 7 128 × 252 =

Division: Use a range of techniques to perform written calculations with whole numbers

4. Calculate the following using any method and check your answers using a calculator:

(a) 3 300 ÷ 25 =

(b) 4 599 ÷ 32 =

(c) 7 028 ÷ 251=

Perform multiple operations on whole numbers with or without brackets

1. Calculate.

(a) 48 ÷ 6 + (4 × 6) × 2 + 6

(b) 3 024 + (60 × 2 ÷ 3)

(c) (5 340 + 160) + 2 – (240 × 2 – 230)

(d) 5 978 ÷ 427 + 51 ÷ 17 × 2

1.3 Ratio and Rate

1. A florist has 7 red roses and 3 white roses in her garden.

FINAL DRAFT COPY

(a) What is the ratio of white roses to red roses and of red roses to white roses?

(b) Write your answer as a common fraction and a percentage

2. Karabo drove a total of 230 km and used 23 litres of petrol. What is the rate of his petrol consumption in kilometres per litre?
3. A tennis player hits 60 aces in 12 matches. Determine the rate of his aces per match.
4. Khaya cycles 21 kilometers per hour. What distance does he cycle in 12 hours?

FINAL DRAFT COPY

UNIT 2

FRACTIONS

2.1 Common Fractions

Addition and Subtraction of common fractions with the same denominators.

1. Calculate each of the following:

(a) $\frac{3}{12} + \frac{5}{12}$

(b) $\frac{7}{8} - \frac{3}{8}$

Addition and Subtraction of mixed numbers with the same denominators.

2. Add:

(a) $9\frac{7}{10} + 5\frac{3}{10}$

(b) $5\frac{2}{9} + 3\frac{5}{9}$

Fractions of whole numbers

3. Calculate how much money is $\frac{1}{5}$ of R200

Recognise, describe and use the equivalence of fractions

4. Look at the following diagram and consider the fraction quarters. It can be written as $\frac{3}{4}$.

(a) Multiply both the numerator and the denominator by 2 to form a new fraction.

(b) Is the new fraction equivalent to $\frac{3}{4}$? You may check on the diagram.

(c) Multiply both the numerator and the denominator by 3 to form a new fraction. Is the new fraction equivalent to $\frac{3}{4}$.

(d) Multiply both the numerator and the denominator by 4 to form a new fraction. Is the new fraction equivalent to $\frac{3}{4}$?

(e) Multiply both the numerator and the denominator by 6 to form a new fraction. Is the new fraction equivalent to $\frac{3}{4}$?

Percentages of whole numbers

5. How much is 80% of each of the following?

(a) R500

(b) R480

(c) R850

(d) R2 400

FINAL DRAFT COPY

Problem solving in context involving common fractions including grouping and equal sharing.

6. Calculate:

(a) Jamie has 16 bottles of juice and needs 3 quarters ($\frac{3}{4}$) of a bottle to fill one glass.

(i) How many quarters of a bottle of juice are there in 16 full bottles?

(ii) How many glasses can he fill with these quarters?

(b) Jamie has 25 bottles of juice and needs $\frac{3}{5}$ of a bottle to fill one glass. How many glasses can he fill?

Problem solving in context involving percentages

7. Solve

(a) Jannie eats a quarter of a watermelon. What percentage of the watermelon is this?

(b) Sibu drinks 75% of the milk in a bottle. What fraction of the milk in the bottle has he drunk?

(c) Building costs of houses increased by 20%. What is the new building cost for a house that previously cost R110 000 to build?

Recognize and use equivalent forms of common fractions (fractions in which one denominator is a multiple of another).

8. Calculate:

(a) $5\frac{1}{9} - 4\frac{2}{3}$

(b) $\frac{13}{20} - \frac{2}{5}$

Recognize equivalence between common fraction and percentage forms of the same number.

9. Find the percentage of the following fractions

(a) $\frac{12}{100}$

(b) $\frac{41}{50}$

FINAL DRAFT COPY

10. Write the following percentages as fractions
    (a) 70%
    (b) 66%

2.2 Decimal Fractions

Compare and order decimal fractions to at least two decimal places

1. Order the following numbers from biggest to smallest. Explain your method.
   (a) 0,8; 0,05; 0,5; 0,15; 0,465; 0,55; 0,75; 0,4; 0,62

2. Below are the results of some of the 2012 London Olympic events. In each case, order them from first to last place. Use the column provided.
   Women: Long jump – Final

NAME	COUNTRY	DISTANCE	PLACE
Anna Nazarova	RUS	6,77m	
Brittney Reese	USA	7,12m	
Elena Sokolova	RUS	7,07m	
Ineta Radevica	LAT	6,88m	
Janay De Loach	USA	6,89m	
Lyudmila Kolchanova	RUS	6,76m	

Place value of digits to at least two decimal places

3. What is the place value of the underlined digits?
   (a) 9**0**0,**0**09
   (b) 4,67**2**

Rounding off decimal fractions to at least 2 decimal places

4. Round off the following decimal fractions to at least two decimal places.
   (a) 365,8760
   (b) 564,234

Addition and subtraction of decimal fractions with at least two decimal places.

5. Calculate the value of the following:

FINAL DRAFT COPY

(a) 8,26 + 5,79 – 2,42

(b) 21,01 – 5,64 – 0,39

(c) 0,78 + 17,64 + 6,65

Multiply decimal fractions by 10 and 100

6. Calculate the value of the following:

		Multiply by 10	Multiply by 100	Multiply by 1000
(a)	82,679			
(b)	640,39			
(c)	0,785			

Solve problems in context involving decimal fractions

7. Alison buys a cooldrink for R5,95; a chocolate for R3,25 and a packet of chips for R4,60. She pays with a R20 note.
   (a) How much did she spend?
   (b) How much change did she get?
8. A tractor uses 11,25 ℓ of fuel in 0,75 hours. How many litres does it use in one hour?

Recognize equivalence between common fraction, decimal fraction and percentage forms of the same number

9. Write the following as common fraction, decimal fraction and percentage forms of the same number

No.	Percentage	Common Fraction	Decimal Fraction
(a)	1,25%		
(b)		$\frac{12}{50}$	
(c)			0,018

FINAL DRAFT COPY

FINAL DRAFT COPY

UNIT 3

ALGEBRAIC LANGUAGE

1. Complete the table below by writing each statement as an algebraic equation.

	Statement	Algebraic expression
(a)	The sum of 6 and a number.	
(b)	A number multiplied by 3 and increased by 4.	
(c)	4 more than a number.	
(d)	The difference between a number and 10	
(e)	The difference between 10 and a number	
(f)	14 less than a number	
(g)	8 less than the product of 5 and a number	
(h)	The sum of a number and 3 multiplied by 7	

(i)	The square root of a number	
(j)	The square of a number	

2. Solve equations by inspection
   a) Determine the value of x if x plus 4 is equals to 9
   b) Solve for x if $2x + 4 = 16$
   c) Solve for x if $x^3 = 27$
   d) Solve for m if $m^2 = 25$
   e) Solve for p if $\frac{p}{3} + 5 = 7$
3. Solve the following equations using trial and improvement

(a) $b - 14 = 6$

b	$b - 14$	Comment
10		
12		
16		
18		
20		

(b) $3x + 5 = 20$

x	$3x + 5$	Comment
2		
3		
4		
5		

(c) $\frac{x}{5} + 9 = 12$

x	$\frac{x}{5} + 9$	Comment
5		
10		
15		

FINAL DRAFT COPY

4. Solve the equations by substitution

(a) $y + 8 = 24$

(b) $z - 9 = 9$

(c) $6x + 35 = 65$

(d) $\frac{x}{4} + 12 = 17$

(e) $x^2 = 36$

5. Identify variables and constants in the table.

No.	Expression	Variable(s)	Constant
(a)	$(2x + y) = 6$		
(b)	$z^3 - 10$		
(c)	$4p^7$		
(d)	$k + 17$		

6. Interpret and write the following rules in words

(a) $\frac{1}{4}w - 4$

(b) $(5k + 11) \times 5$

(c) $10n - 5$

(d) $(z \times 6) + 14$

(e) $x^3 + \sqrt{q}$

7. Substitute to determine the numerical value of the following expressions.

No.	x	-1	0	2
(a)	$x - 4$			
(b)	$5x + 4 + 3x - 2$			
(c)	$(4x + 7)(3x - 2)$			

FINAL DRAFT COPY

FINAL DRAFT COPY

UNIT 4

NUMERIC AND GEOMETRIC PATTERNS

Investigate, extend and describe numeric and geometric patterns

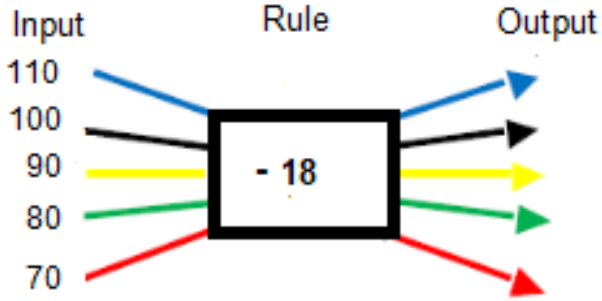
1. What are the next three numbers in each of these sequences?
   a) 4; 7; 10; 13; 16;
   b) 22; 18; 14; 10; 6;
   c) 37; 42; 47; 52; 57;
   d) 7; 14; 28; 56; 112;

2. Identify whether the sequence has constant difference or constant ratio
   (a) 60; 65; 70; 75
   (b) 200; 175; 150; 125
   (c) 625; 125; 25; 5

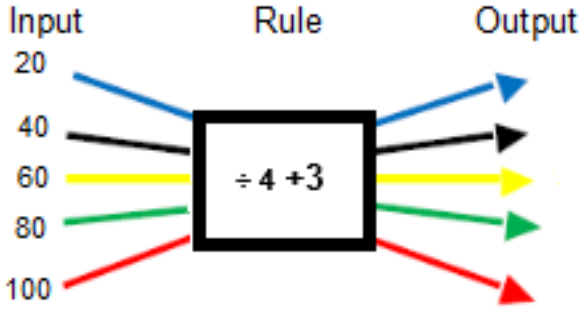
(d) 6; 18; 54; 162

3. Complete the flow diagrams

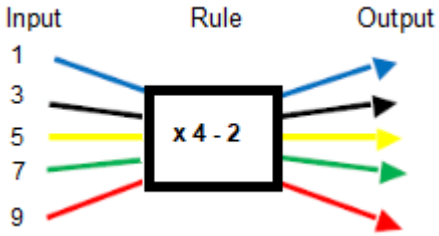
(a)



(b)



(c)



4. Complete the following tables using the given rules:

(a) The rule for calculating area of a square for a garden is “Multiply the length of a side of the square by itself”

Length of the side	4	6		10	
Area of square	16		64		144

(b) Add 37 to the previous term to get the next one

FINAL DRAFT COPY

Position(n)	1	2	3	4	5	6
Term	3	40				

(c) Subtract 3 from the previous term to get the next one

Position(n)	1	2	3	4	5	6
Term	70					

(d) Multiply by 4

Position(n)	1	2	3	4	5	6
Term						

5. Create your own numeric pattern, represent it in words, in a flow diagram and in a table.

6. Study the pattern in the table below:

Position(n)	1	2	3	4	5	6
Term	2	5	10	17	26	37

a) What is the value of the 7^{th} term?

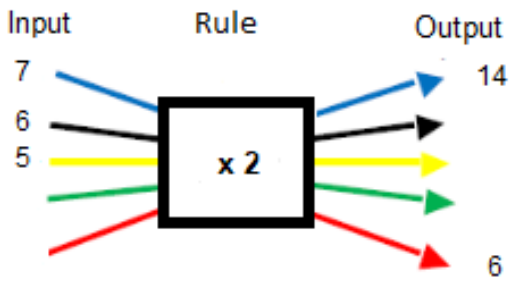
b) Describe the rule in words to determine value of any term (n^{th} term) in the sequence.

c) Write down the formula to determine the value of any term in the sequence.

d) Use your formula to determine the 10^{th} term of the pattern

7. Complete the flow diagram and use it to determine the equivalence

(a) Flow diagram



(b) Represent the relationship in the flow diagram verbally

FINAL DRAFT COPY

(c) Represent in a table

Input(n)	7	6	5		
Output	14				6

(d) Write the general rule or formula

8. Study the following geometric patterns and draw figure 5.

(a)

Figure 1 Figure 2 Figure 3 Figure 4 Figure 5

(b)

Figure 1 Figure 2 Figure 3 Figure 4 Figure 5

(c)

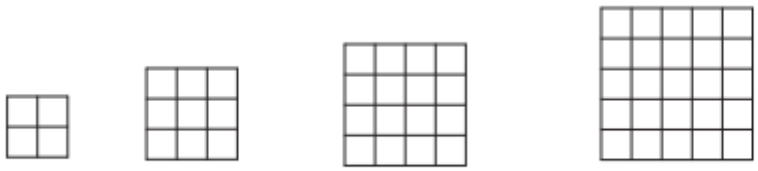


Figure 1 Figure 2 Figure 3 Figure 4 Figure 5

9. Mpho builds a pattern using circles, study her pattern and answer the questions that follows:

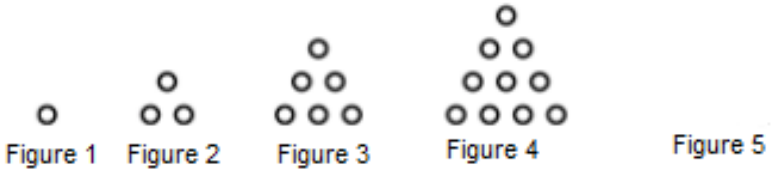


Figure 1 Figure 2 Figure 3 Figure 4 Figure 5

(a) Draw the fifth figure:

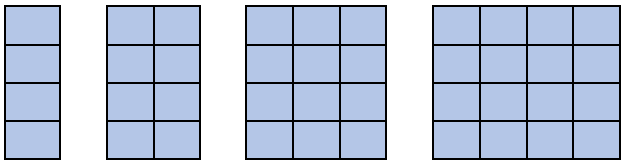
(b) How many circles are added in each figure?

(c) Write down the formula to determine the number of matchsticks in each figure.

(d) Use your formula to determine the number of matchsticks in the 12th figure.

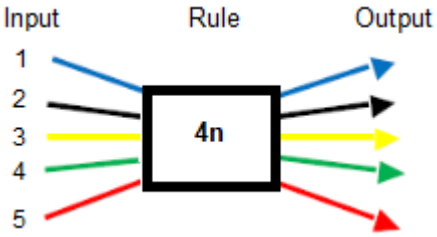
10. Study the diagram below. Represent the equivalence.

(a) Draw the fifth figure.



F1 F2 F3 F4 F5

(b) Complete the flow diagram.



(c) Use the information from flow diagram to complete the table

Figure					
No of squares					

(d) The constant difference?

(e) The pattern in own words

FINAL DRAFT COPY

FINAL DRAFT COPY

UNIT 5

MEASUREMENT

5.1 Length

Solve problems in contexts involving length, mass and capacity/volume

1. Joseph is a designer and has 10 rolls of material each roll is 6 metres each.
   (a) How much material does he have altogether?
   (b) If he uses 4,5 metres to design one dress, how many metres will he need to design 13 dresses?
   (c) How much material will be left?
2. In a marathon, Buli ran 12 300 metres and Lethu ran 12,3 kilometres. Who ran the shortest distance, explain your answer?

5.2 Mass

Solve the following problems.

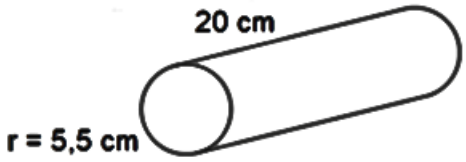
1. A farmer sells oranges for R24,99 a bag. Each bag weighs 7kg and he supplies twelve retail stores with 36 bags each. How many kilograms will be received by each retail store and how much will he be paid altogether?
2. A retail store has a sale for cornflakes. They sell 4 times 1,2 kilograms for R150.The original price of the same cornflakes is R24,99 for a 750-gram box. Which is the best buy and why?
3. Sendi and Lerato are twins, Sendi's birth weight was 2,15 kg and Lerato's was 728 g. Who had more weight? Explain your answer.
4. My mother bought a 10 kg bag of rice. She gave her sister one quarters of the rice. How much rice did she give her sister, write your answer in grams?

5.3 Capacity and Volume

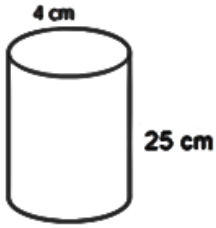
Solve the following problems.

1. A cube – shaped water reservoir has a side length of 12 m.
   (a) What is the capacity of the reservoir? Write your answer in litres.
   (b) If the reservoir is quarter full of water, what does it hold in kilolitres?
2. In a bucket there are 30 litres of juice. How many learners can get 375 millilitres of juice?
3. The monthly water use for a certain family is as follows:
   76 000 millilitres to wash the dishes; 150 litres to bath; 26,5 litres for cooking; $12\frac{1}{2}$ litres for gardening and 12 000 millilitres for drinking. How much water does the family use monthly? Write your answer in litres.
4. For my sister's 16^{th} birthday party, my aunt made 5 000 millilitres of cocktail juice. She used 375 millilitres of mango juice and 4,5 litres of strawberry juice. How many millilitres of granadilla juice has she added to get the 5 000 millilitres of cocktail?
5. A motor car uses 15,50 ℓ of fuel in 0,25 hours. How many litres does it use in one hour?
6. Calculate the volume and capacity of the following cylinders. Use π = 3,14 and round off all the answers to two decimal places.
   Volume of a cylinder = Area of base × $h = \pi r^2 \times h$
   (a)

FINAL DRAFT COPY



(b)

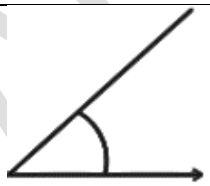
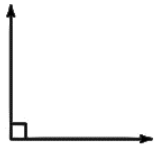


7. Use a formula to calculate the volume of water that will fill each cube. Convert your answer into millimetres, litres and kilolitres.

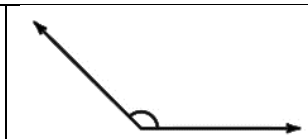
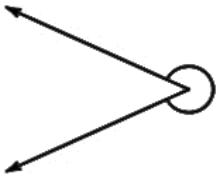
UNIT 6

CONSTRUCTION OF GEOMETRIC FIGURES

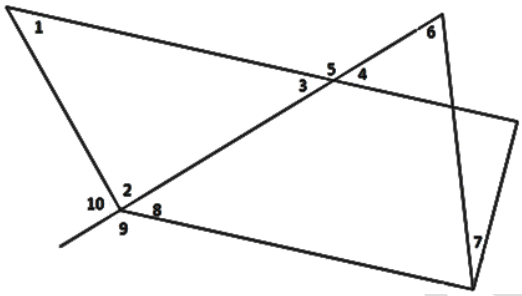
1. Measure the following angles using your protractor.

No.	Angle	Size
(a)		
(b)		
(c)		

FINAL DRAFT COPY

(d)		
(e)		
(f)		

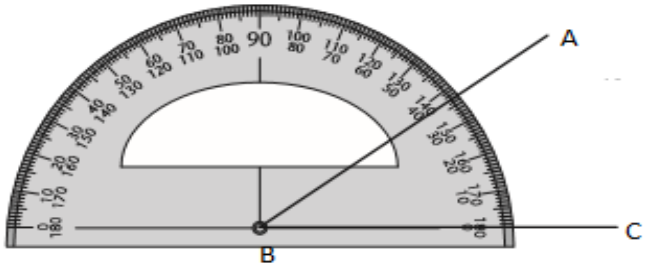
2. Measure all the numbered angles in the following figure.



Angle	1	2	3	4	5	6	7	8	9	10
Size	47°	88°	45°	45°	135°	65°	21°	44°	136°	92°

3. Construct the following angles.

(a) $A\hat{B}C$ =37°using a protractor



(b) $X\hat{Y}Z = 110°$ using a protractor

(c) $U\hat{V}W = 240°$ using a protractor

(d) $D\hat{E}F = 60°$ using a compass

(e) $K\hat{L}M = 120°$

4. Construct a circle with a:

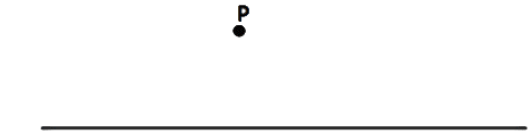
FINAL DRAFT COPY

(a) Radius of 4 cm

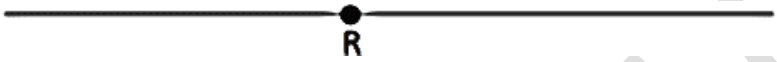
(b) Diameter of 7 cm

5. Construct perpendicular lines.

   (a) From point P to the line segment



   (b) Point R in the line segment below.



6. Construct parallel lines with line MN parallel to line OP.

7. Draw the perpendicular line QR which is parallel to lines MN and OP using a protractor and a ruler. Given line MN and point Q draw a parallel line OP to MN that passes through point Q.

8. Use a protractor to construct the following.

   a) Equilateral triangle

   b) Kite, the angle P must be 52˚ and the one at R must be 23˚

FINAL ART COPY

UNIT 7

GEOMETRY OF STRAIGHT LINES

5. Define the following terms
   (a) Line segment
   (b) Ray
   (c) Straight line
   (d) Parallel lines
   (e) Perpendicular lines

6. Highlight the parallel lines in this picture

FINAL DRAFT COPY

Picture A

Picture B

Picture C

Picture D

7. Identify the parallel and the perpendicular lines in the pictures.

Picture A

Picture B

Picture C

8. Circle the intersecting lines which forms vertically opposite angles in the pictures.

Picture A

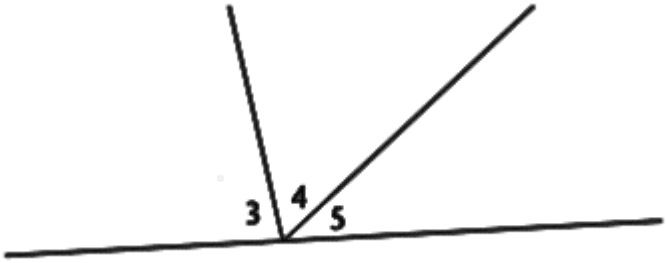
Picture B

9. Use a protractor to measure the sizes of all the angles in each figure. Write your answers in each figure.

FINAL DRAFT COPY

(a)

(b)

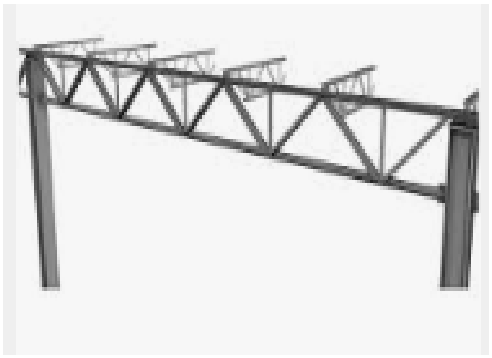


(c) Use your answers in (a) and (b) above to say if the angles are supplementary or complementary

$\hat{1} + \hat{2} =$

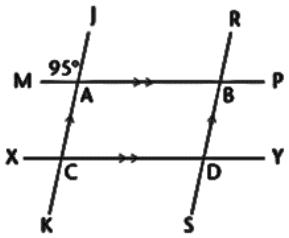
$\hat{3} + \hat{4} + \hat{5} =$

10. Study the diagram of girders below, identify and label the complementary and supplementary angles

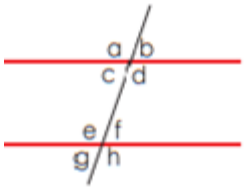


11. Find the sizes of all the angles. (Can you see two transversals and two sets of parallel lines?)

FINAL DRAFT COPY



12. Identify pairs of equal angles and write their special name.



UNIT 8

GEOMETRY OF 2-D SHAPES

1. Identify triangles in terms of their sides and angles

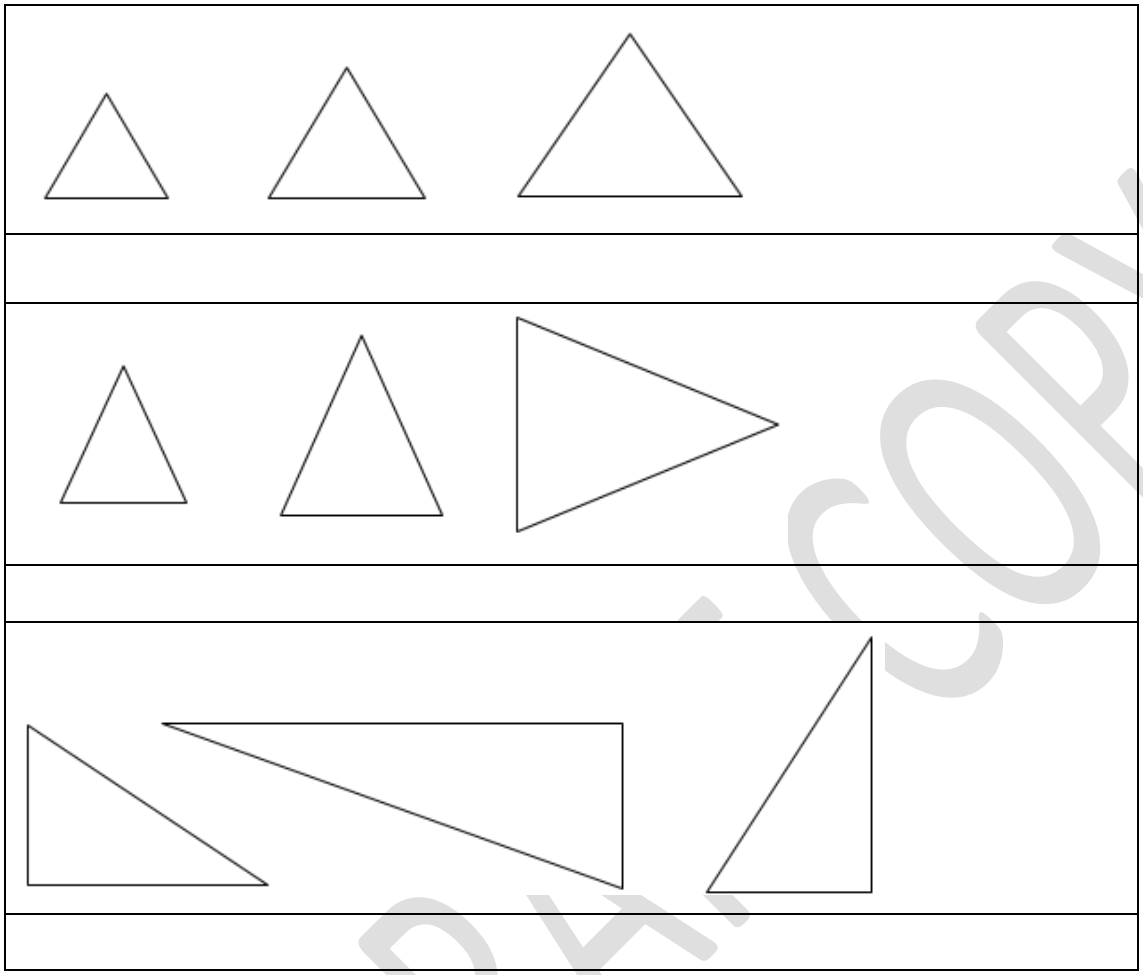
   (a) equilateral triangles

   (b) isosceles triangles

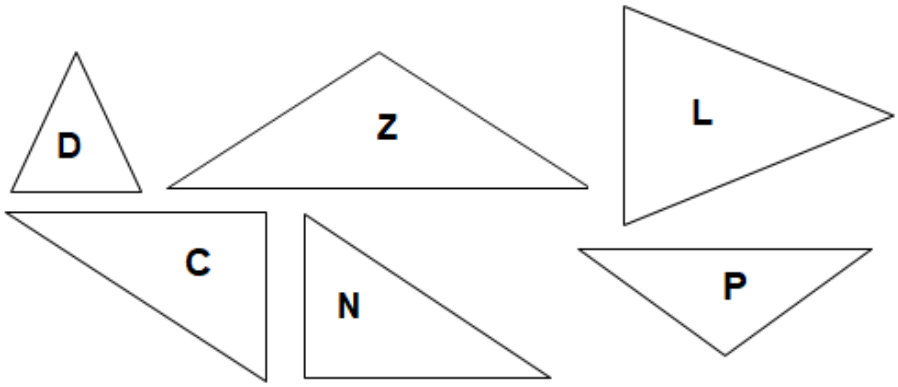
   (c) right-angled triangles

   Measure the **sides** of each triangle below. Classify the triangles according to the measurements you got. Motivate why you classified them the way you did.

FINAL DRAFT COPY

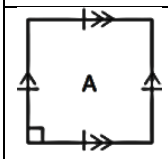
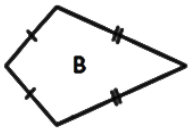
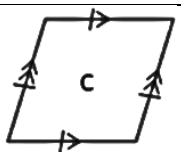
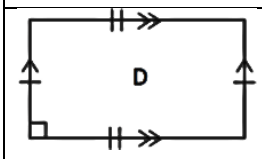
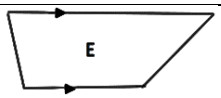
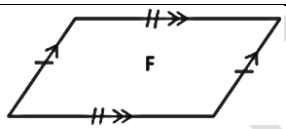


2. Measure the **angles** of each triangle below. Classify the triangles according to the measurements you got. Motivate why you classified them the way you did.



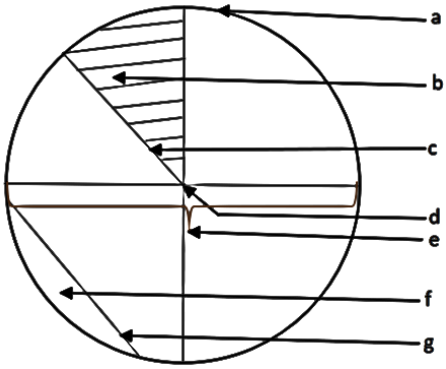
FINAL DRAFT COPY

3. Study the figures below and give their properties in terms of the sides and angles.

Figure	Name	Properties in terms of:	
		Sides	Angles
			
			
			
			
			
			

Describing and naming parts of a circle.

4. Label the parts of the circle marked (a-g) below:



FINAL DRAFT COPY

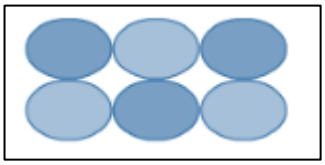
FINAL DRAFT COPY

UNIT 9

TRANSFORMATIONS

Use transformations to make composite shapes and tessellations

1. Which shapes have been used in the patterns below? Are these patterns tessellations? In which way have the shapes been tessellated? (translation, reflection or rotation)

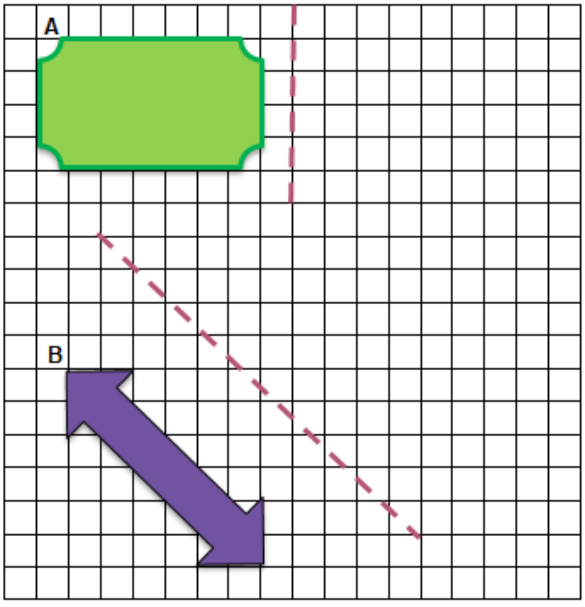
(a)	(b) 
(c)	(d)

Reflection, Transaction and Rotation

2. Perform the following transformations on a grid.

(a) Reflect shape A along the vertical line.

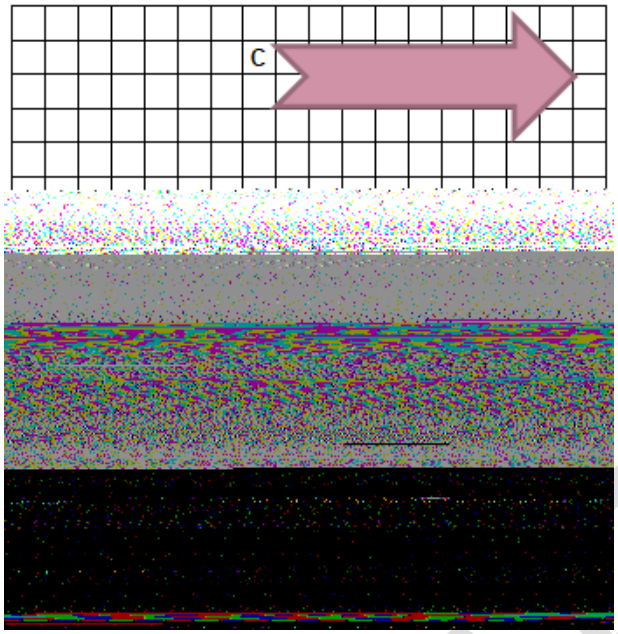
Reflect shape B along the diagonal line.



FINAL DRAFT COPY

(b) Translate shape C, 4 units to the left and 3 units' down

Translate shape D, 2 units up and 6 units to the right.



(c) Rotate shape E 180° anticlockwise

Rotate shape F 90° clockwise

Enlargement and reduction

1. On grid paper, enlarge the rectangle below with a scale factor of 3

FINAL DRAFT COPY